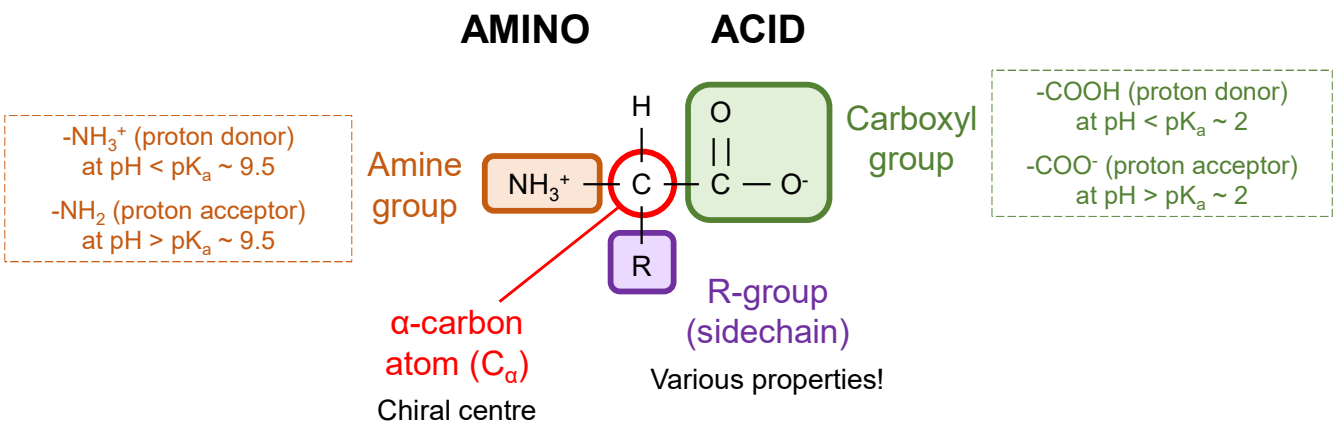
BIOC0001 – Proteins and Enzymes

Amino Acids

Dr. Renée Vancraenenbroeck
October 26, 2026

The same small set of building blocks – **amino acids** – is used to produce proteins with different structures and functions. Here, we will look at some properties of amino acids.

Common Features of Amino Acids



An **α-amino acid** has a central carbon atom, the **α-carbon atom** ($C_α$) which carries:

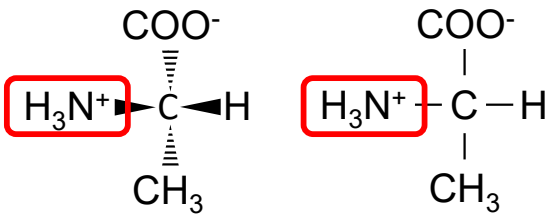
- an **amine group** (-NH_2) – which can be protonated,
- a **carboxyl group** (-COOH) – which can be deprotonated,
- a hydrogen atom,
- and a **sidechain or R-group**.

The side chain can have various functional groups and gives each amino acid distinct physical properties influencing a protein's structure and function.

Of note, proline is an exception and contains a ring structure including the $C_α$ atom and amine group.

Stereoisomerism: L and D

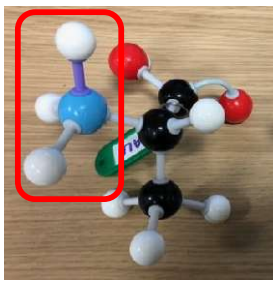
In proteins!



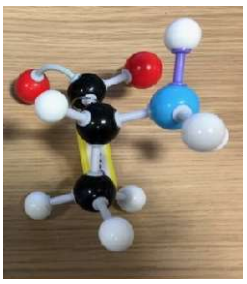
Dashed-wedged line structure

Fisher projection

L-ALANINE



D-ALANINE



MIRROR

More help? Chiral vs achiral | Stereochemistry | Organic chemistry | Khan Academy - YouTube

The α-carbon atom is a **chiral centre** because it is bonded to four different groups. Amino acids have thus two possible **stereoisomers**, called the **L and D** stereoisomers.

- They are a class of stereoisomers called **enantiomers** because they are non-superimposable mirror images of each other.
- They are **optically active**. Historically, the similar *l* and *d* designations were used for *l*evorotatory (rotating light to the left) and *d*extrorotatory (rotating light to the right). However, not all L-amino acids are levorotatory!

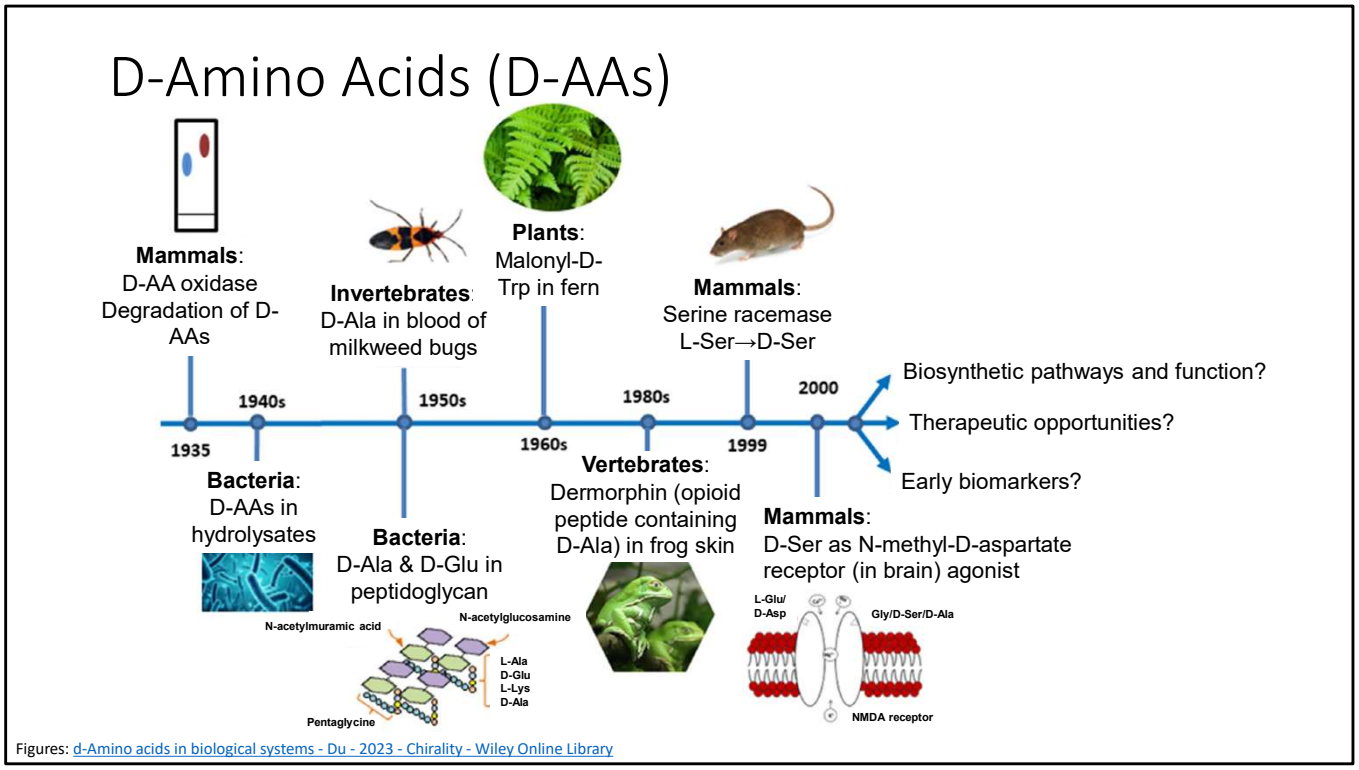
To determine whether an amino acid is the L or D stereoisomer:

1. Draw the molecule as a Fischer projection with the carboxyl group on top and sidechain on the bottom.
2. If the amino group is located on the ***l***eft side of the carbon chain, the molecule is ***L***. If the α-amine group is on the right side, the compound is D.

In nature, the amino acids **in proteins** are **L stereoisomers**. Because of the presence of L-amino acids, α-helices are usually right-handed.

Of note, glycine has a hydrogen atom as R-group and is not chiral.

For more help, see the "Chiral vs achiral" YouTube movie from Khan Academy Organic Chemistry. Available through https://youtu.be/yChPRfm4yvg.



Figures: d-Amino acids in biological systems - Du - 2023 - Chirality - Wiley Online Library

D-amino acids (D-AAs) were once thought to exist only in **bacteria**. However, D-AAs are now known to occur across **plants, invertebrates, and vertebrates – including mammals**.

In mammals, **free D-AAs** such as D-Ser, have emerged as important **signalling molecules** in various organs, especially the brain. Their involvement in multiple disorders suggests they could serve as **early biomarkers**, and that studying them further may reveal new **therapeutic opportunities**.

D-AA-containing peptides (DAACPs) have can arise in several ways:

- **Spontaneous isomerization**. Certain amino acids, especially **aspartic acid**, can convert non-enzymatically into their D-form during ageing. This can disrupt protein structure and function. D-Asp has been detected in **αA-crystallin** from cataract patients' eye lenses. D-Asp also occurs in **β-amyloid** in the brains of Alzheimer's patients.
- **Enzyme-catalysed post-translational modifications (PTMs)**. **Dermorphin**, an opioid peptide from frog skin, contains D-Ala even though its gene encodes L-Ala. Peptide isomerase activity has been found in frog skin secretions.
- **Non-ribosomal peptide synthesis**. **Bacteria** incorporate D-Ala and D-Glu into their **cell walls** using specific enzymes. Many antibiotics with D-AAs, such as **penicillin** (contains D-Val) and **gramicidin S** (contains D-Phe) are made step by step by specialised non-ribosomal peptide synthetases.

Twenty Common Amino Acids

Amino acid	Abbreviation 3-letter	Abbreviation 1-letter	Amino acid	Abbreviation 3-letter	Abbreviation 1-letter
Alanine	Ala	A	Leucine *	Leu	L
Arginine +	Arg	R	Lysine *	Lys	K
Asparagine +	Asn	N	Methionine *	Met	M
Aspartic acid	Asp	D	Phenylalanine *	Phe	F
Cysteine	Cys	C	Proline +	Pro	P
Glutamic acid	Glu	E	Serine +	Ser	S
Glutamine +	Gln	Q	Threonine *	Thr	T
Glycine +	Gly	G	Tryptophan *	Trp	W
Histidine *	His	H	Tyrosine +	Tyr	Y
Isoleucine *	Ile	I	Valine *	Val	V

* = Essential, needs to be obtained from the diet

+ = Conditionally essential, essential under certain circumstances (illness, injury, stress ...)

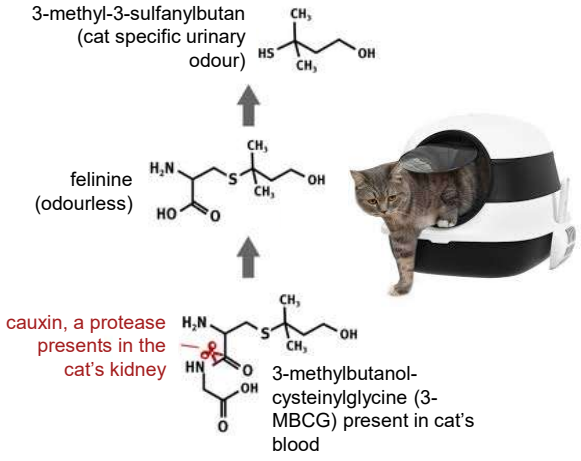
Twenty different amino acids are **common**ly found **in proteins**. They have been assigned **3-letter and 1-letter abbreviations**, which are used as shorthand.

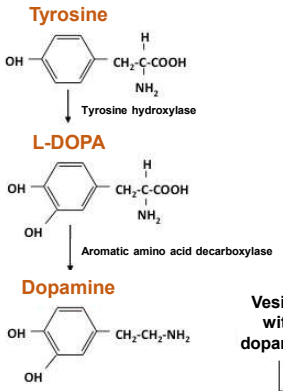
In some cases, the names are derived from the source from which they were first identified: asparagine – asparagus; glutamate – wheat gluten; tyrosine – cheese (from the Greek *tyros,* "cheese") ... The single letter codes were introduced by Margaret Oakley Dayhoff using first letters (*e.g.* Alanine, Threonine ...), phonetically suggestive letters (*e.g.* aRginine – R, Phenylalanine – F ...), and related letters (*e.g.* asparagiNe contains N ...).

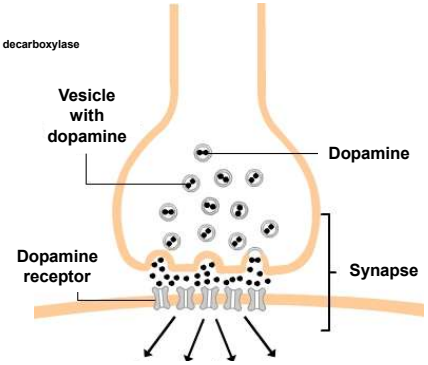
The twenty common amino acids can be sorted into two groups: **essential and non-essential**. Non-essential amino acids are those which the human body is capable of synthesising, whereas essential amino acids must be obtained from the diet.

Conditionally essential amino acids are nonessential amino acids that become essential under specific conditions. For example, in **phenylketonuria**, an inherited metabolic disorder, the body cannot properly convert the amino acid phenylalanine into tyrosine due to a deficiency in the enzyme phenylalanine hydroxylase. This deficiency leads to high levels of phenylalanine and low levels of tyrosine. The main treatment is a lifetime diet with very limited intake of foods with phenylalanine.

Non-proteinogenic Amino Acids (npAAs) Felinine and L-DOPA







Figures: Pathophysiology of L-Dopa Induced Dyskinesia — Changes in D1/D3 Receptors and Their Signaling Pathway | IntechOpen and How cats produce their distinctive smell | RIKEN

The 20 common protein amino acids (plus selenocysteine and pyrrolysine) represent a small proportion (~2%) of the total number of amino acids known to exist in nature. **Non-proteinogenic amino acids (npAAs)** are abundant. They are not incorporated into proteins but still play important biological roles, such as **neurotransmission (L-DOPA)** and **sociochemical communication (felinine)**.

L-DOPA is derived from the amino acid tyrosine and converted to dopamine in neurons. Dopamine is crucial for signalling in the brain. L-DOPA is used in treating Parkinson's disease.

Felinine is found in the urine of cats. It is derived from 3-MBCG with the help of cauxin. It is processed and responsible for the characteristic cat urinary odour.

Properties

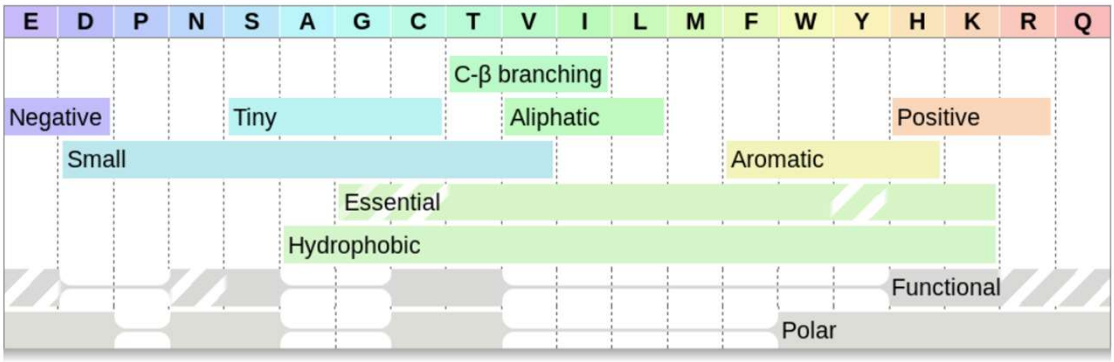


Figure: A New Diagram for Amino Acids: User Study Comparing Rainbow Boxes to Venn/Euler Diagram | IEEE Conference Publication | IEEE Xplore

The 20 common amino acid side chains have different properties: **hydrophobic; aliphatic; aromatic; polar; positively charged; negatively charged; small; tiny; C-β branching**.

It is important to *know the alphabet, structure, and properties of each amino acid by heart to talk sensibly and quickly with your colleagues* about proteins and enzymes. Imagine you have a conversation with a colleague about COVID-19 mutations. When they mention *e.g.* the E484K mutation in the spike protein, you will be able to immediately (without looking up the alphabet, structure, and properties!) visualise the small, negatively charged glutamic acid at position 484 being replaced by a large, positively charged lysine. A drastic change in the sequence. How does this mutation influence structure and function? Food for discussions which are fuelled by experiments!

To facilitate these types of discussions (and many others!), it is critical that life sciences students learn the alphabet, structure, and properties of the amino acids. To learn them, the best method is memorization.

P.S. There are differences between textbooks / internet sources regarding classifications, it can be *confusing*!

Hydrophobic and Polar!

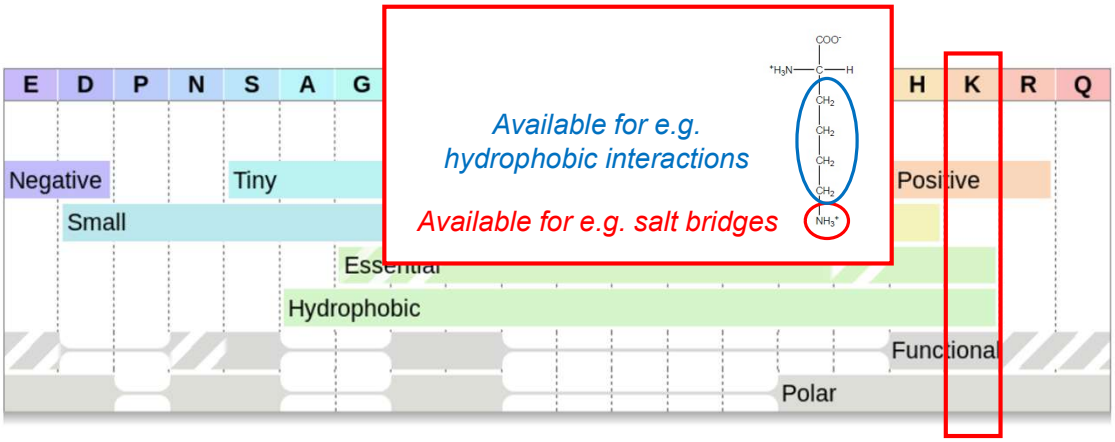


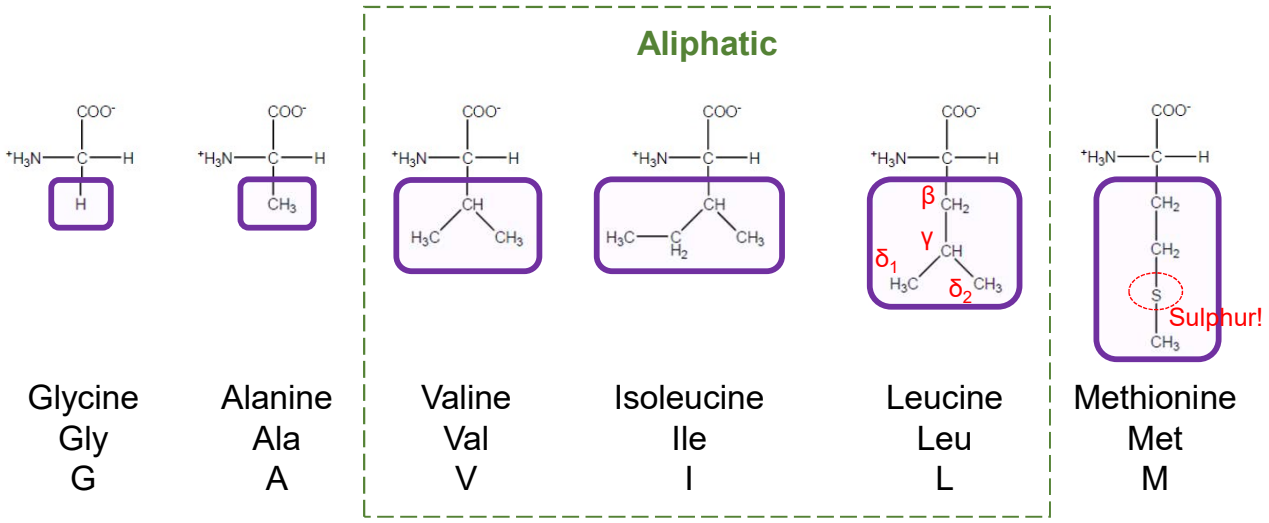
Figure: A New Diagram for Amino Acids: User Study Comparing Rainbow Boxes to Venn/Euler Diagram | IEEE Conference Publication | IEEE Xplore

Some residues, *e.g.* Cys, Thr, Tyr, Trp, Lys and His, have both polar and nonpolar character, making them **amphipathic**. For example:

- Lysine carries a charge, which can lead to salt bridges; but lysine is also often involved in hydrophobic interactions due to its long hydrophobic part. Have a look at the Russell lab pages for some statistics about lysine *(optional)*: http://www.russelllab.org/aas/Lys.html
- Tyrosine can participate in both: hydrophobic (with its aromatic side chain) and polar (because of the tyrosine hydroxyl group) interactions. Have a look at the Russell lab pages for some statistics about tyrosine *(optional)*: http://www.russelllab.org/aas/Tyr.html

P.S. Some scientists also classify amino acids with nonpolar side chains at physiological pH as amphipathic as they contain polar groups, *i.e.* the protonated α-amine group ($–NH_3^+$), and deprotonated α-carboxyl group ($-COO^-$).

Hydrophobic



Alanine, valine, leucine, isoleucine and methionine are usually found within proteins, stabilizing protein structure by means of **hydrophobic interactions**.

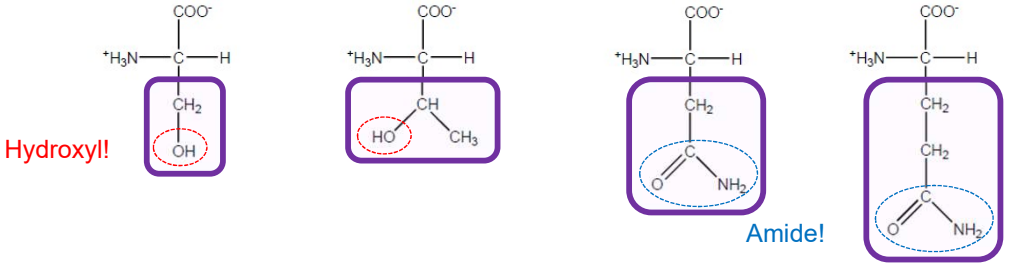
Glycine is formally nonpolar but as its side chain is a hydrogen atom, it makes no real contribution to hydrophobic interactions. As its side chain is tiny, it usually resides in parts of protein structures that are forbidden to all other amino acids (*e.g.* turns in structures).

Methionine is one of the two sulphur-containing amino acids, the other one is cysteine. Methionine's sulphur is connected to a methyl group which makes the roles that methionine can play in protein function (*e.g.* metal coordination) much more limited than cysteine.

Valine, isoleucine and leucine are classified as **aliphatic** because they are not polar, not charged, and not aromatic. Alanine is not classified in the aliphatic group as it is very small and Met is not classified in the aliphatic group as it contains a sulphur atom.

The additional carbons in an R group are commonly designated **β, Y ...** proceeding out from the α-carbon. For branched amino acid side chains, equivalent carbons are given numbers after the Greek letters, *e.g.* leucine has δ1 and δ2 carbons.

Polar, Uncharged



Serine	Threonine	Asparagine	Glutamine
Ser	Thr	Asn	Gln
S	T	N	Q

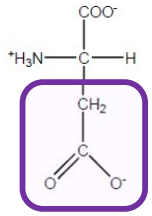
The R groups of these amino acids are more soluble in water because they contain functional groups that form **hydrogen bonds**.

Serine and threonine have hydroxyl groups and **asparagine and glutamine** amide groups.

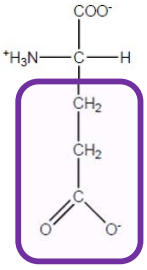
These amino acids generally prefer to be on the surface of proteins, exposed to an aqueous environment.

Polar, Charged

Negatively charged; Acidic

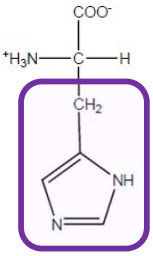


Aspartic acid
Asp
D

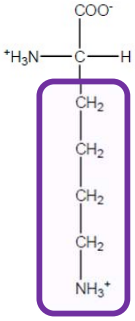


Glutamic acid
Glu
E

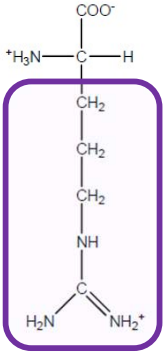
Positively charged; Basic



Histidine
His
H



Lysine
Lys
K



Arginine
Arg
R

Charged and polar, these amino acids prefer to be on the surface of proteins, exposed to an aqueous environment. When buried within the protein, they are frequently involved in **salt bridges**.

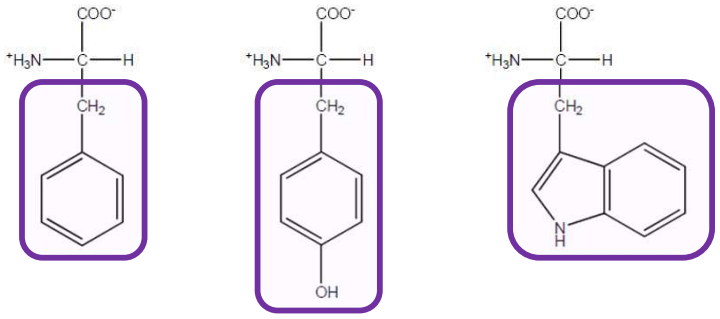
Negatively Charged (Acidic) R Groups

Amino acids in which the R groups have a net negative charge at physiological pH are **aspartic acid and glutamic acid**, both have a second carboxyl group.

Positively Charged (Basic) R Groups

Amino acids in which the R groups have significant positive charge at physiological pH are **lysine**, with a second amino group; **arginine**, with a guanidino group; and **histidine**, with an imidazole group. Histidine has a pK_a near to that of physiological pH. In many enzyme-catalyzed reactions, histidine facilitates the reaction by serving as a proton donor / acceptor.

Aromatic



Phenylalanine
Phe
F

Tyrosine
Tyr
Y

Tryptophan
Trp
W

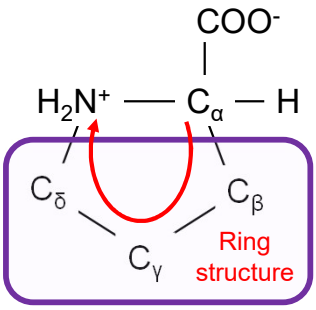
Phenylalanine, tyrosine, and tryptophan, with their aromatic side chains, are relatively nonpolar. All can participate in hydrophobic interactions.

Tyrosine and tryptophan are significantly more polar than phenylalanine, because of the tyrosine hydroxyl group and the nitrogen of the tryptophan indole ring.

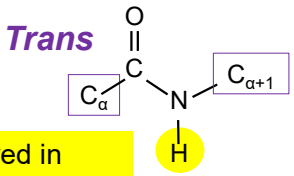
Tryptophan and tyrosine, and to a much lesser extent phenylalanine, **absorb ultraviolet light**. This accounts for the characteristic strong absorbance of light by most proteins at a wavelength of 280 nm.

P.S. All amino acids absorb at 190-220 nm due to the peptide bond.

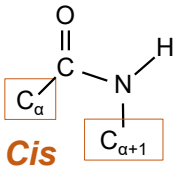
Special: Proline



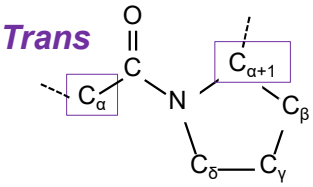
Proline
Pro
P

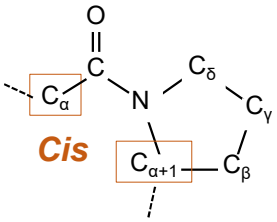


Involved in secondary structure hydrogen bonding



~ 0.05 %





~ 6.5 %

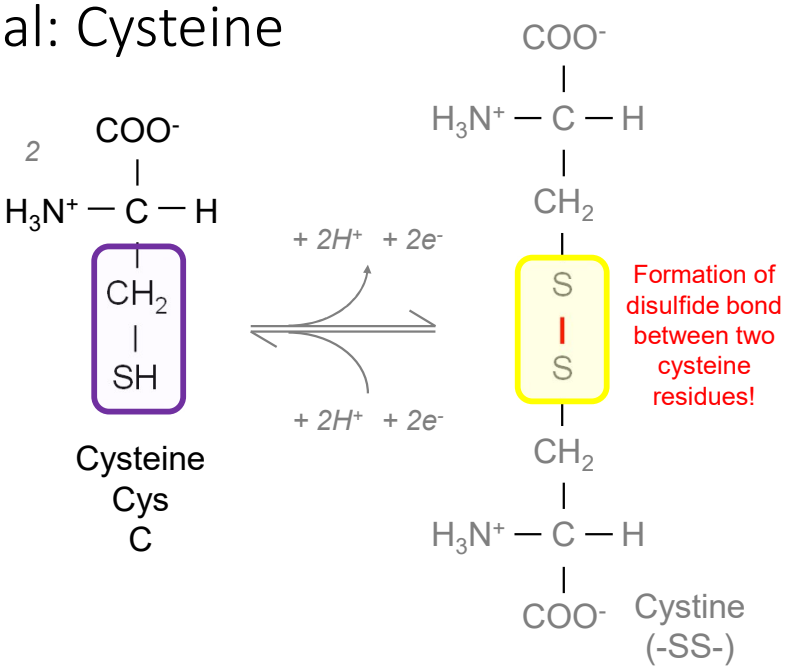
Proline has a distinctive **cyclic structure**: a five-membered nitrogen-containing ring. It also contains an **NH_2^+** rather than an NH_3^+ group in its isolated form at physiological pH.

Proline's side chain is connected to the protein backbone twice, this ring structure holds proline residues in a **rigid conformation**, reducing the structural flexibility of polypeptide regions containing proline.

Peptide bonds between amino acid residues other than proline are all in the *trans* configuration. When proline is involved, about 6% of the bonds are in the ***cis* configuration**.

Proline can often be found in **turn**s in protein structures. The preference for turn structure means that prolines are often found on the protein surface.

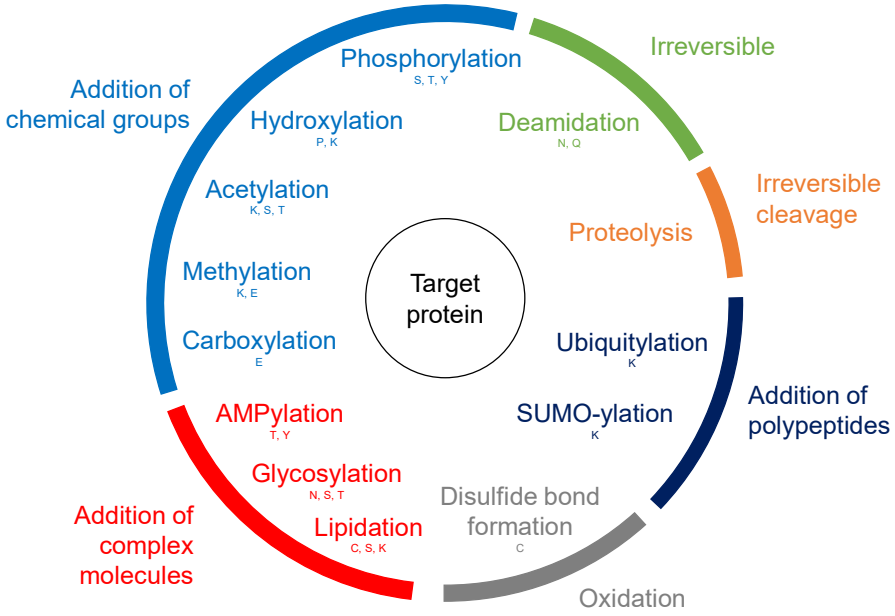
Special: Cysteine



Cysteine is readily oxidized to form a covalently linked dimeric amino acid called **cystine,** in which two cysteine molecules or residues are joined by a **disulphide bond**. The disulphide-linked residues are strongly hydrophobic.

Disulphide bonds play a special role in the structures of many proteins by forming covalent links between parts of a protein molecule or between two different polypeptide chains.

Post-Translational Modifications (PTMs)



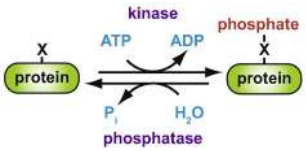
Amino acids already incorporated into a polypeptide can be modified – reversibly or irreversibly. **Post-translational modifications (PTMs)** are catalyzed by specific enzymes and target specific residues. They alter the properties of the amino acid, which can alter the structure and function of the protein, including changes in enzyme activity, interaction with other molecules, cellular localisation, and stability.

Common reversible PTMs include **phosphorylation (on S, T, Y residues), hydroxylation (P, K), acetylation (K, S, T), methylation (K, E), carboxylation (E), AMPylation (T, Y), N- or O-glycosylation (N, S, T), lipidation (C, S, K), disulphide bond formation (C), ubiquitylation (K), and SUMO-ylation (K)**. **Proteolysis** and **deamidation (N, Q)** are examples of irreversible PTMs.

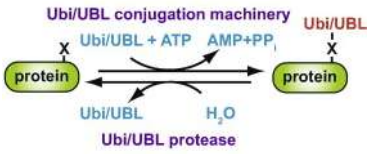
Of note, I only listed some common amino acid modifications. But many more exist! See https://www.uniprot.org/help/mod_res and https://www.uniprot.org/help/post-translational_modification and https://www.uniprot.org/help/carbohyd for a more detailed list *(optional)*.

Post-Translational Modifications (PTMs) *(cont.)*

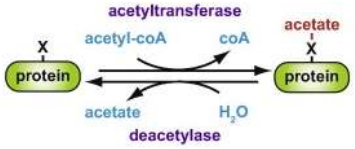
Phosphorylation (S, T, Y)
E.g. receptor phosphorylation



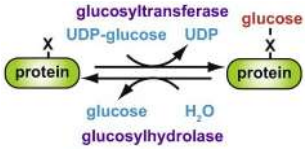
Ubiquitylation (K)
E.g. protein degradation



Acetylation (K, S, T)
E.g. histone acetylation



N- and O-glycosylation (N, S, T)
E.g. erythropoietin



Proteolysis
E.g. prothrombin

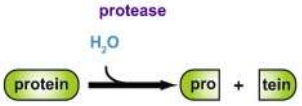
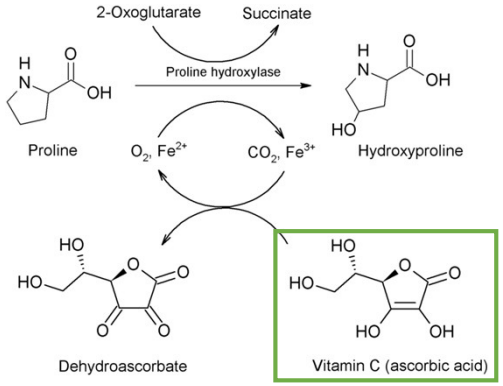


Figure: Post-translational modifications in host cells during bacterial infection - ScienceDirect

Examples include:

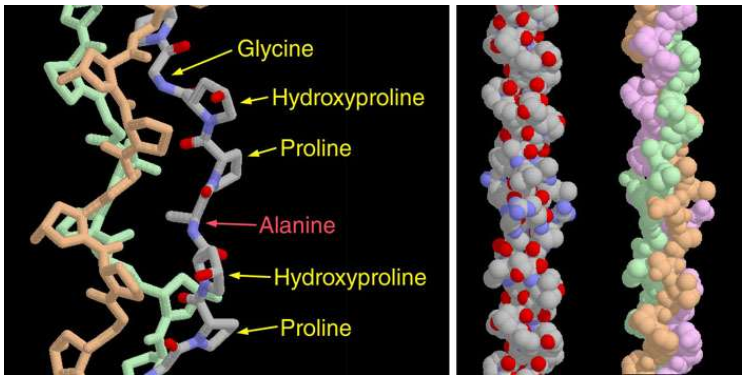
- **Receptor phosphorylation** impacts how receptors interact with other molecules and respond to stimuli (*e.g.* the insulin receptor). Kinases catalyse phosphorylation and phosphatases catalyse dephosphorylation.
- **Histone acetylation**. Histone acetyl transferases transfer acetyl groups from acetyl-CoA to lysine residues on histones, neutralizing their positive charge. This weakens the interaction between histones and DNA, leading to a less condensed and more accessible chromatin structure. Histone deacetylases remove acetyl groups from lysine residues and cause chromatin to compact.
- **Erythropoietin glycosylation**. The blood-stimulating hormone erythropoietin has N- and O-linked oligosaccharides contributing to its function and stability.
- **Ubiquitylation** involves the addition of a small protein called ubiquitin on to other proteins. This involves a large family of proteins, the E2 and E3 ligases. Deubiquitinating enzymes reverse this process. Ubiquitylation is most associated with **protein degradation**.
- **Proteases** enzymatically hydrolyze peptide bonds in substrate proteins, resulting in a widespread, irreversible PTM of the protein's structure and biological function. A famous example is **prothrombin** which is cleaved at two sites when it is converted to active thrombin. Prothrombin is rich in **γ-carboxyglutamate** (another PTM!), which can bind to Ca^{2+} ions, crucial for the activation of prothrombin to thrombin.

PTM: Hydroxyproline and Collagen



Scurvy in case of vitamin C deficiency!

Triple helix, coloured by atom and chain



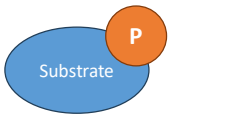
Alanine distorts the triple helix! Only Gly fits!

Figures: analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/mas.21600 and PDB-101: Molecule of the Month: Collagen

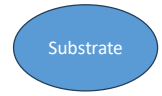
Collagens are proteins that form **fibrous** structures and are abundant in the body: present in skin, blood vessel walls, bones, teeth, and more. Each collagen molecule consists of three polypeptide chains with many repeating **Gly-X-Y** sequences, where X is often **proline** and Y is often **hydroxyproline**. The three chains coil together into a tight **triple helix**. Every third residue passes through the crowded centre of the helix, where only Gly can fit. The bulky, rigid proline and hydroxyproline residues give the entire structure its strength and stability.

Hydroxyproline is generated by post-translational modification of proline by the enzyme **proline hydroxylase**, which requires **vitamin C (ascorbic acid)** for activity. Without vitamin C, hydroxyproline levels drop, and collagen fibres cannot form properly. This leads to **scurvy**, a disease characterized by fragile blood vessels, skin lesions, poor wound healing, and, if untreated, death. Historically, scurvy was a major cause of mortality among sailors deprived of fresh food during long sea voyages.

Salmonella and PTMs



SptP
tyrosine
phosphatase



Cytoskeleton dynamics &
release of inflammatory substances

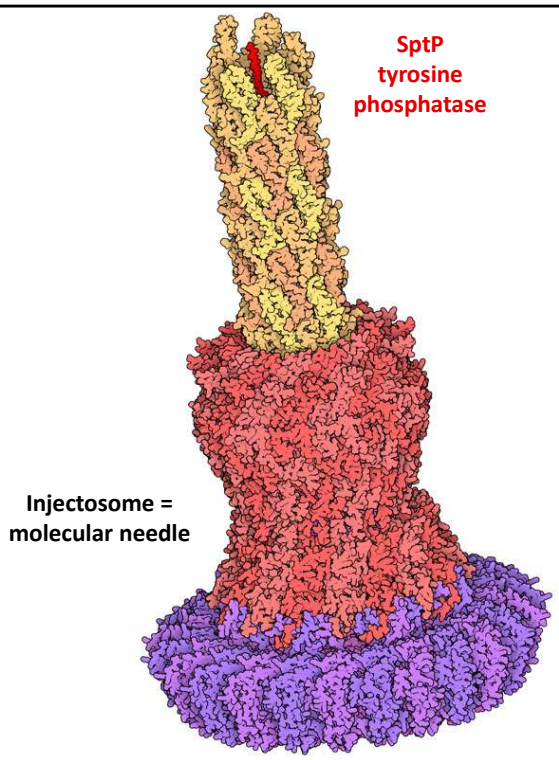


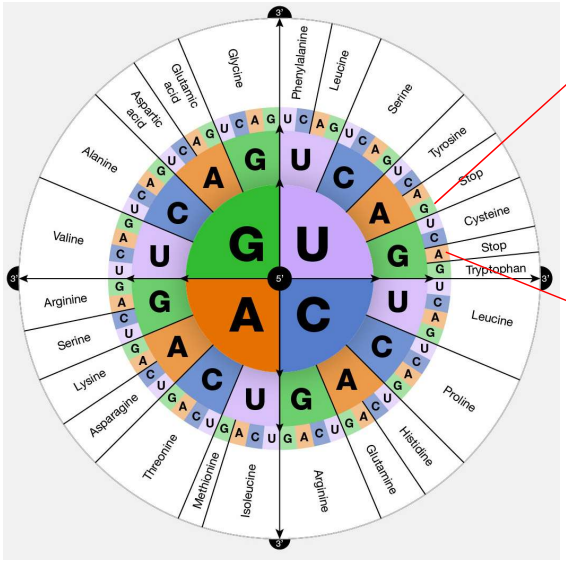
Figure: PDB-101: Molecule of the Month: Injectisome

Salmonella bacteria inject via the **injectisome** a cocktail of **effector proteins** into cells to disable their defences, triggering bacterial uptake. Many of them interrupt the action of cellular proteins by mediating the PTMs of host proteins.

For instance, the effector protein **SptP** is an enzyme that removes phosphates from tyrosines on cellular proteins. Its dephosphorylation of these substrates prevents cytoskeleton dynamics and the release of inflammatory substances from mast cells.

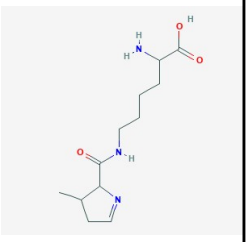
Have a look at https://youtu.be/0ueUfLXWojA for an animation of the injectosome!

Not PTMs: Selenocysteine and Pyrrolysine



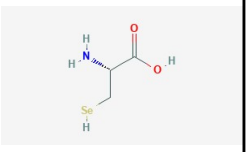
Pyrrolysine (Pyl, O) = 22nd amino acid

- Found in a small number of methanogenic archaea and bacteria
- Synthesized from Lys by three enzymes
- Encoded to a stop (UAG) codon with the help of a unique pyrrolysyl-tRNA synthetase (PylRS) along with its tRNA ($tRNA^{Pyl}$)



Selenocysteine (Sec, U) = 21st amino acid

- Found in selenoproteins in archaea, bacteria and eukaryotes
- Synthesized from Ser on its $tRNA^{Sec}$
- A specialized elongation factor binds to Sec-$tRNA^{Sec}$ and a selenocysteine insertion sequence downstream of the stop (UGA) codon



Figures: Genetic Code and PubChem

Two amino acids are **rare** and are known as the 21st and 22nd amino acids: **selenocysteine (Sec, U) and pyrrolysine (Pyl, O)**. It was long thought that they were the result of PTMs. However, Sec and Pyl are encoded by UGA and UAG codons, respectively, which normally serve as stop signals.

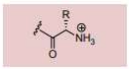
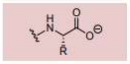
Sec is typically found in the active sites of a small number of selenoproteins in archaea, bacteria and eukaryotes. Pyl is found in a small number of methanogenic archaea and bacteria.

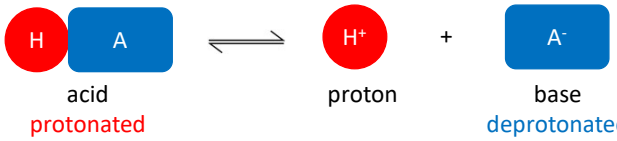
Selenocysteine (Sec) and Pyrrolysine (Pyl) have special mechanisms to be synthesised and incorporated into a protein:

- Pyl – synthesised from Lys by three enzymes – is ligated directly to $tRNA^{Pyl}$ and inserted into proteins in response to UAG codons.
- Sec is synthesized from Ser on its cognate tRNA ($tRNA^{Sec}$), by one enzyme in bacteria and two in archaea and eukaryotes. During translation, Sec-$tRNA^{Sec}$ is delivered to the ribosome by a specific elongation factor (via another protein in archaea and eukaryotes) that requires a characteristic stem-loop structure in the mRNA to actively recode an in-frame UGA from stop codon to Sec sense codon.

Selenocysteine and pyrrolysine are powerful examples of the versatility in the genetic code. They also show how apparently similar phenomena can have different mechanisms!

Amino Acids Act as Acids and Bases

Terminus	pK_a	Low pH; protonated state; (conjugated) acid; proton donor	High pH; deprotonated state; (conjugated) base; proton acceptor
N-terminus	~ 9.5 ~ 8 in proteins		
C-terminus	~ 2 ~ 3 in proteins		



$$K_a = \frac{[H^+] \times [A^-]}{[HA]}$$

$$pH = pK_a + \log_{10}\frac{[A^-]}{[HA]} \quad \text{with} \quad pH = -\log_{10}[H^+] \quad \text{and} \quad pK_a = -\log_{10} K_a$$

– The lower the pK_a, the stronger the acid *(the more easily it dissociates)!*

– $when\ pH = pK_a \leftrightarrow [A^-] = [HA];\ when\ pH < pK_a \leftrightarrow [HA] > [A^-];\ when\ pH > pK_a \leftrightarrow [A^-] > [HA]$

– $when\ pH = pK_a - 1\ ;\ [A^-] = 0.1\ [HA];\ when\ pH = pK_a + 1\ ;\ [A^-] = 10\ [HA]$ …

More help? https://biochemistry-teaching-resources.ucl.ac.uk/biochemistry/bianimations_5/proteins/tutorials/titratn/titratn_iframes.php?co=BIOC0001

Because amino acids have acidic and basic groups, their charge can change depending on the environment.

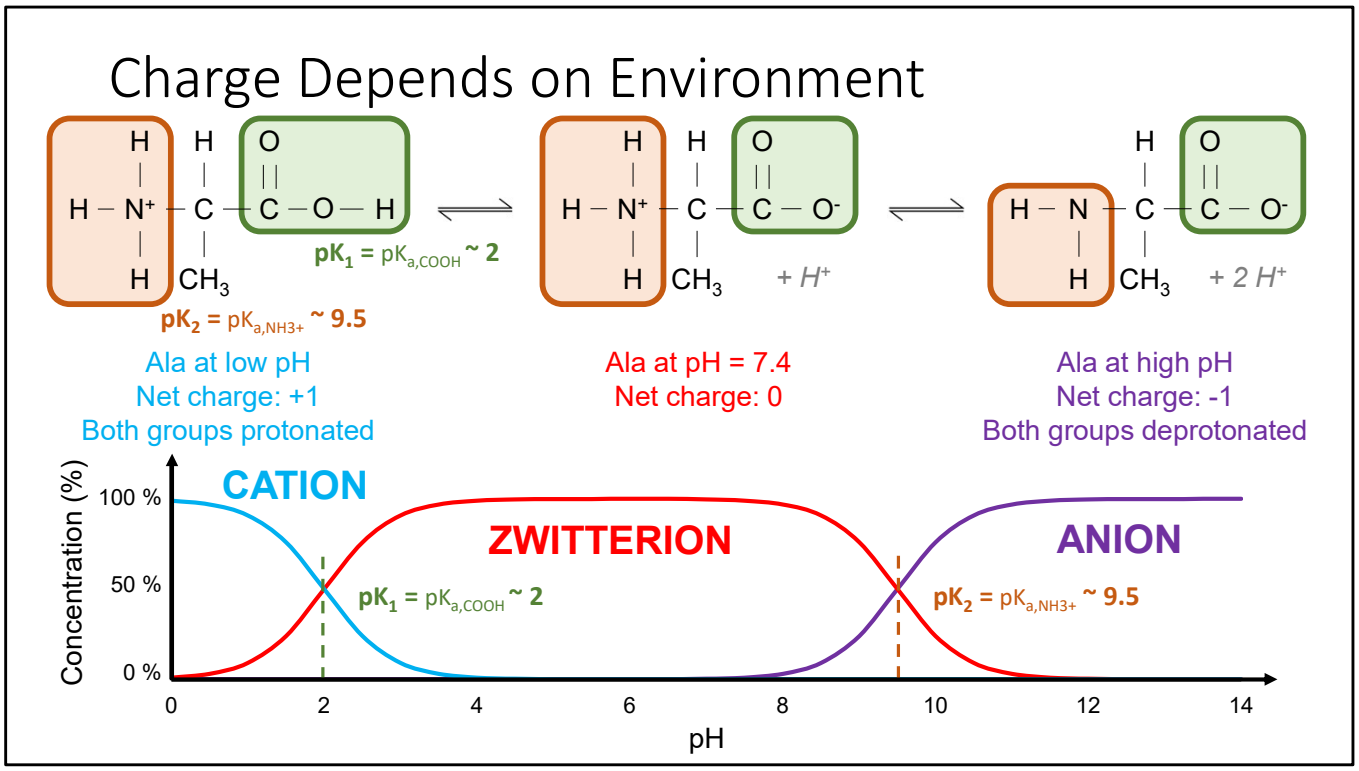
The figure shows the ionization state of the **amine and carboxyl groups** that predominantly exists at physiological pH, *i.e.* pH 7; highlighted in red if the indicated ionization form is at least 100 times more abundant than the other form. These ratios were calculated using the **Henderson-Hasselbalch equation**, a formula that relates the pH of an aqueous acid solution to its acid dissociation constant K_a.

Important to remember:

- According to the Bronsted-Lowry definition, acids donate protons and bases accept protons. Every acid has a conjugate base and every base a conjugate acid.
- The pK_a describes the strength of an acid: the lower the pK_a, the stronger the acid and the more easily it dissociates.
- The value of $[A^-]/[HA]$ is dependent on the value of the pH and pK_a:
  - When pH < pK_a; we have more HA.
  - When pH = pK_a; we have equal amounts of HA and A^-, *i.e.* the acid is 50% dissociated.
  - When pH > pK_a; we have more A^-.
- Because we work with logarithmic terms, for every unit change in the pH to pK_a ratio, a tenfold change occurs in $[A^-]/[HA]$.

Refresh and test your knowledge on **acid, base, pH, pK_a, the Henderson-Hasselbalch equation, and titrations** (seen in A-level chemistry and/or university chemistry modules) by going to https://biochemistry-teaching-resources.ucl.ac.uk/biochemistry/bianimations_5/proteins/tutorials/titratn/titratn_iframes.php?co=BIOC0001, an interactive tool to review their concepts and apply them to the ionisation of amino acids.

- Log in with your UCL username and password.
- Your answers are given a score. At the end of the animation, your total score is shown on the screen and you are sent an email for your records. Please keep track of these emails because we are not able to answer individual queries about your animation results. If you have not seen the final score on the screen or received an email, then you have not completed the animation. Scores do not count towards your final mark in BIOC0001!



Let us have a detailed look at alanine. It has amine and carboxyl groups capable of ionizing.

- At low pH, when alanine is fully protonated, it is a diprotic acid with net charge +1. It is a **cation**. It has two groups, the –COOH group and the $–NH_3^+$ group, that can donate protons. Alanine has two acid dissociation constants: ~ 2 for the α-carboxyl (COOH) group and ~ 9.5 for the α-amino (NH_3^+) group. **The lower the pK_a of an acid, the stronger the acid and the more easily it "*gives up*" its proton.** So, for alanine, the α-carboxyl dissociates first ($pK_1 = pK_{a,COOH} \sim 2$), and the α-amino dissociates second ($pK_2 = pK_{a,NH3+} \sim 9.5$).
- When alanine is dissolved in water (at pH ≈ 7), it exists in solution as the dipolar ion, or **zwitterion with no net charge**. A zwitterion can act as either an acid (proton donor) or a base (proton acceptor).
- At high pH, when alanine is fully deprotonated, it is a diprotic base with net charge -1. It is an **anion**. It has two groups, the $–COO^-$ group and the $–NH_2$ group, that can accept protons.

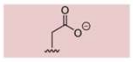
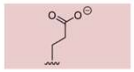
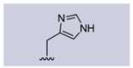
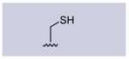
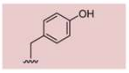
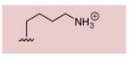
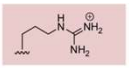
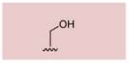
The pH dependence of the ionization states – the cation, zwitterion, and anion – of alanine is shown as well. It was calculated using the **Henderson-Hasselbalch equation**. The two groups, the –COOH group and the $–NH_3^+$ group, were treated as independent as their pK_a values are far from each other.

- When pK_a and pH are equal, exactly half of the acid has dissociated ([A^-

]/[HA] = 1). At pH 2, we have 50 % cation and 50 % zwitterion. At pH 9.5, we have 50 % zwitterion and 50 % anion.

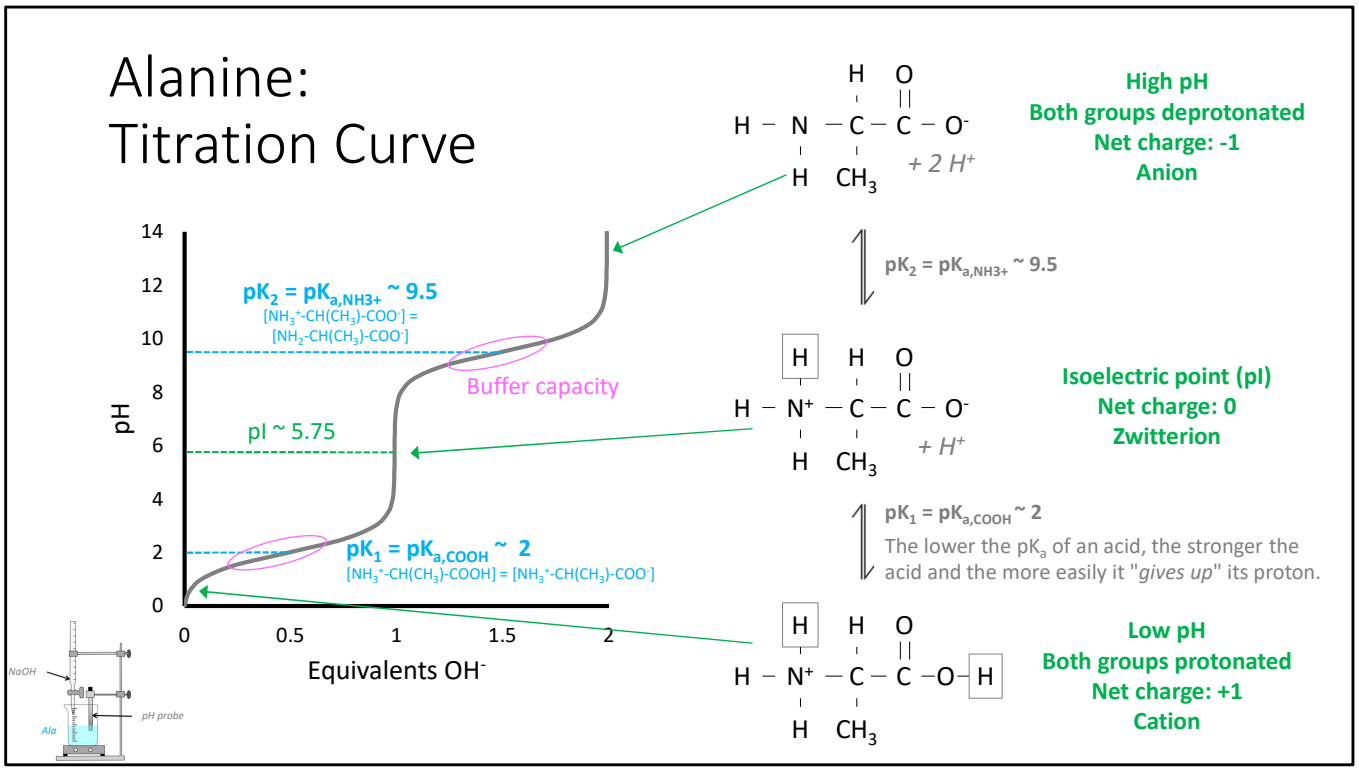
- When the pH is pK_a - 1, over 90 % exists as acid ($[A^-]/[HA] = 0.1$). At pH 1 we have over 90 % cation and less than 10 % zwitterion. At pH 8.5, we have over 90 % zwitterion and less than 10 % anion.
- When the pH is pK_a + 1, over 90 % exists as base ($[A^-]/[HA] = 10$). At pH 3, we have over 90 % zwitterion and less than 10 % cation. At pH 10.5, we have over 90 % anion and less than 10 % zwitterion.

Amino Acids Act as Acids and Bases *(cont.)*

Amino acid	pK_a of sidechain	Low pH; protonated state; (conjugated) acid; proton donor	High pH; deprotonated state; (conjugated) base; proton acceptor
Aspartic acid	~ 4		
Glutamic acid	~ 4		
Histidine	~ 6		
Cysteine	~ 8.5		
Tyrosine	~ 10.5		
Lysine	~ 10.5		
Arginine	~ 12.5		
Serine	~ 13		

The figure shows the ionization state of the **side chain groups** that predominantly exists at physiological pH, *i.e.* pH 7; highlighted in red if the indicated ionization form is at least 100 times more abundant than the other form and highlighted in purple if not.

- We classify aspartic acid and glutamic acid as acids due to their negative charge at pH 7.
- We classify lysine and arginine as bases due to their positive charge at pH 7.
- Histidine is a special one as its pK_a is close to 7. At physiological pH, we have a mixture of its positively charged, protonated form and its neutrally charged, deprotonated form. We classify it as a base.
- Cysteine, tyrosine, serine, and threonine (the latter is not shown in the figure) are neutral at pH 7 and we do not generally classify them as acidic or basic amino acids.



Here, the **titration curve** of alanine, *i.e.* an amino acid with two ionisable groups, with a strong base (*e.g.* NaOH) is shown. The curve displays what happens to alanine as you change the pH by adding the strong base. The curve is calculated using the Henderson-Hasselbalch equation.

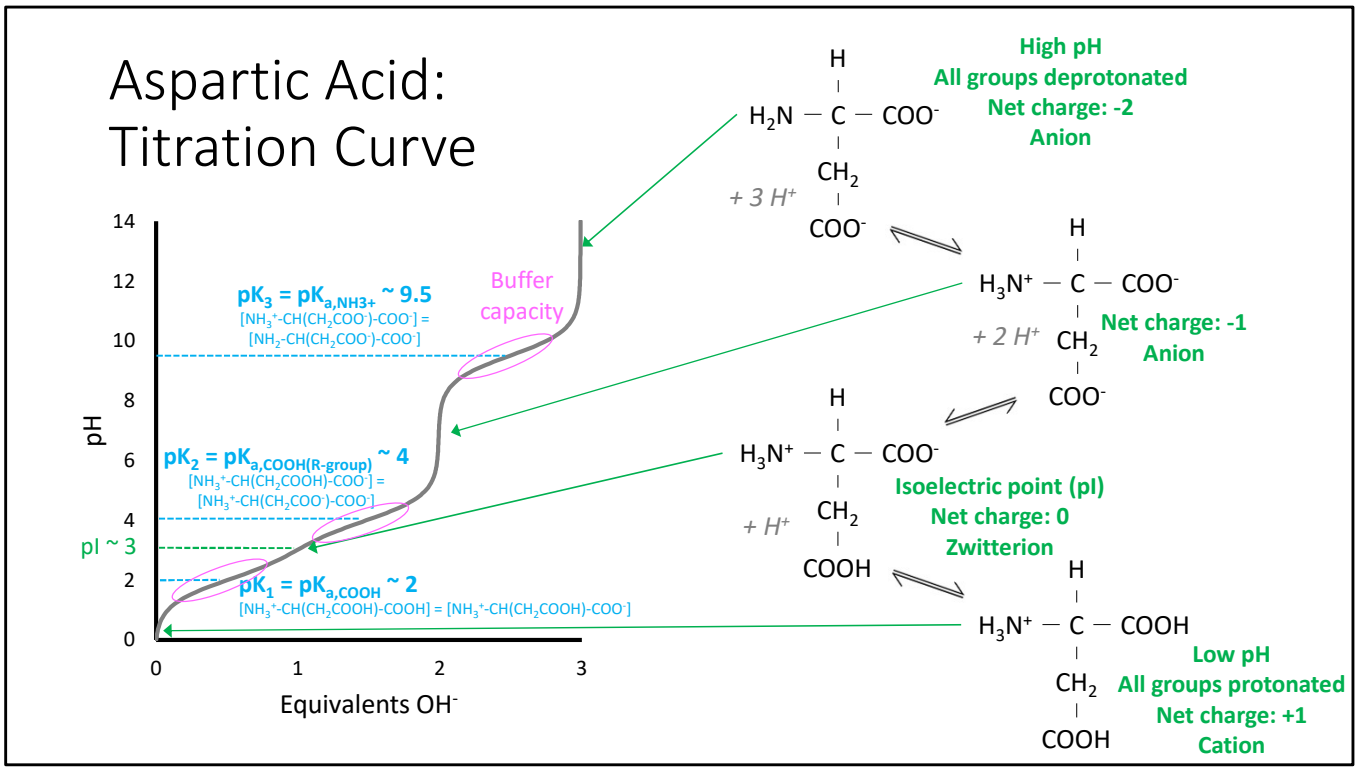
The plot has distinct features, corresponding to deprotonation of two different groups on alanine.

- At **low pH**, when alanine is **fully protonated**, it is a diprotic acid with **net charge +1**. It is a **cation**. It has two groups, the –COOH group and the $–NH_3^+$ group, that can donate protons.
- At **high pH**, when alanine is **fully deprotonated**, it is a diprotic base with **net charge -1**. It is an **anion**. It has two groups, the $–COO^-$ group and the $–NH_2$ group, that can accept protons.
- A midpoint can be seen at pH 5.75. Here, the removal of the first proton is essentially complete and removal of the second has just begun. This point is known as the **isoelectric point (pI)** and is loosely defined as the pH where the amino acid carries **no net charge** and is a **zwitterion**. A zwitterion can act as either an acid (proton donor) or a base (proton acceptor). When alanine is dissolved in water (at pH ≈ 7), it exists in solution as the dipolar ion, or zwitterion with no net charge.
- Alanine has two acid dissociation constants.
    - At the midpoint in the first stage of the titration, in which the -COOH

group loses its proton, equimolar concentrations of the proton-donor (-COOH) and proton-acceptor ($-COO^-$) species are present. The pH is equal to the pK_a of the protonated group being titrated; here $\mathbf{pK_1} \sim 2$, with 1 representing -COOH.

- At the midpoint in the second and last stage of the titration, in which the $-NH_3^+$ group loses its proton, equimolar concentrations of the proton-donor ($-NH_3^+$) and proton-acceptor ($-NH_2$) species are present. The pH is equal to the pK_a of the protonated group being titrated; here $\mathbf{pK_2} \sim 9.5$, with 2 representing $-NH_3^+$.
- Remember, the lower the pK_a of an acid, the stronger the acid and the more easily it "*gives up*" its proton.

- To change the pH level near the pK_a value of an acid, the dissociation status of the acid must be changed significantly, which requires using an extremely large amount of strong base. The ability of a substance to maintain the pH of such solutions is referred to as its **buffer capacity**. Alanine has two regions of **buffering power**, one around pH 2 ± 1 and one around pH 9.5 ± 1.



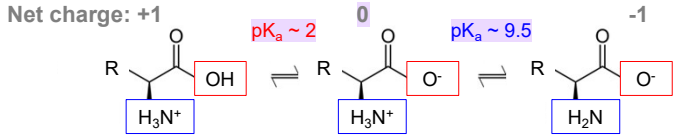
In this example, the **titration curve** of aspartic acid, *i.e.* an amino acid with three ionisable groups, with a strong base (*e.g.* NaOH) is shown.

This titration curve is more complex, with three *(compared to two)* stages corresponding to the three possible ionization steps. Aspartic acid has three pK_a values: 1 representing –COOH, 2 representing the R-group –COOH, and 3 representing $-NH_3^+$. The additional stage for the titration of the ionizable R group merges to some extent with the other two.

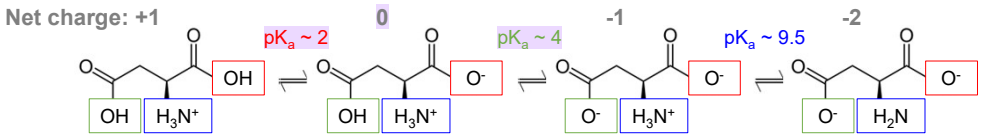
Iso-electric Point

The isoelectric point (pI) is the **pH** at which the amino acid or peptide has **no net charge** and is **a zwitterion**.

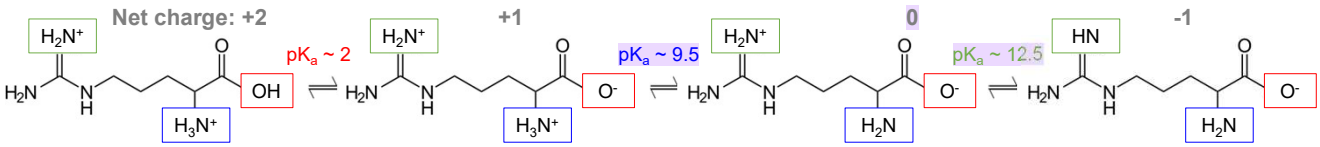
It can be determined by **averaging the pK_a values of the two groups which straddle the zwitterion.**



Gly, Ala, Val ... have two ionisable groups: $pI = \frac{pK_{a,COOH} + p\ \ _{a,NH3}}{2} = \frac{2+9.5}{2} = 5.75$



E.g. Asp has three ionisable groups: $pI = \frac{pK_{a,COOH(C\alpha)} + pK_{a,COOH(sidechain)}}{2} = \frac{2+4}{2} = 3$



E.g. Arg has three ionisable groups: $pI = \frac{pK_{a,NH3+(C\alpha)} + pK_{a,NH2+(sidechain)}}{2} = \frac{9.5+12.5}{2} = 11$

More help? How To Calculate The Isoelectric Point of Amino Acids and Zwitterions – YouTube

The characteristic pH at which the net charge is 0 is called the **isoelectric point** or **pI**.

For alanine, which has **no ionizable group in its sidechain**, the isoelectric point is simply the arithmetic mean of the two pK_a values: the -COOH pK_a and the $-NH_3^+$ pK_a *(see the titration curve of alanine from earlier)*.

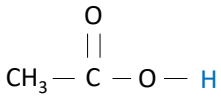
For aspartic **acid**, with an ionisable group in its sidechain, the isoelectric point can be determined by averaging the pK_a values of the two groups which are closest to and straddle the isoelectric point: the pK_a values for the carboxylic acids (pK_1, the -COOH pK_a and pK_2, the R-group pK_a) *(see the titration curve of aspartic acid from earlier)*.

The isoelectric point of lysine, a **basic** residue, can be determined by averaging the pK_a values of the two groups which are closest to and straddle the isoelectric point: the pK_a values for the amino groups (pK_2, the $-NH_3^+$ pK_a and pK_3, the R-group pK_a) *(it is helpful to sketch this out yourself!)*.

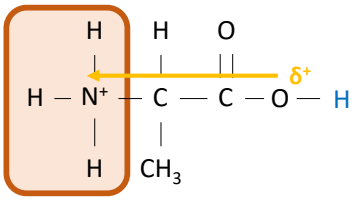
For more help, see the "How to calculate the isoelectric point of amino acids and zwitterions" YouTube movie from The Organic Chemistry Tutor, available https://www.youtube.com/watch?v=Gkb4it5nOuc.

pK_a Values Depend on the Environment

Acetic acid, carboxyl group
Departing proton, pK_a ~ 4.5

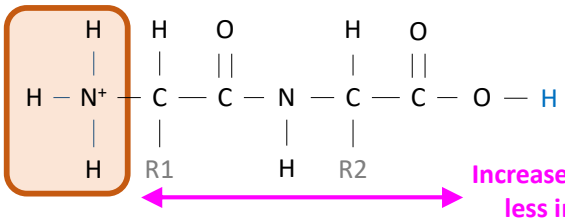


Amino acid, carboxyl group
Departing proton, pK_a ~ 2



Positively charged amino group, electron-withdrawing

Protein, carboxyl group
Departing proton , pK_a ~ 3



Increased distance, less influence

We saw easy-to-remember approximations for amino acid pK_a values. Be aware though that **the pK_a of any functional group is greatly affected by its chemical environment.**

The pK_a of the **carboxyl group of an amino acid** is lower than that of the carboxyl group of **acetic acid** (pK_a ~ 4.5). The carboxyl group of an amino acid is more acidic and "*gives up*" its proton more easily. The increased acidity of the α-carboxyl group of an amino acid reflects the electron-withdrawing inductive effect of the positively charged α-amino group (this group is protonated at the pH that the carboxyl group will lose its proton!). Also, in the case for diprotic amino acids such as alanine, the opposite charges on the resulting zwitterion are stabilizing.

In **proteins**, the only available amino and carboxyl groups are on the N- and C-terminus respectively; the other amino and carboxyl groups were used to make polypeptide bonds and no longer exist. The pK_a values of these N-and C-terminal groups are different to free amino acids as an oppositely charged group is no longer linked to the *α*-carbon.

pK_a Values Depend on the Environment *(cont.)*

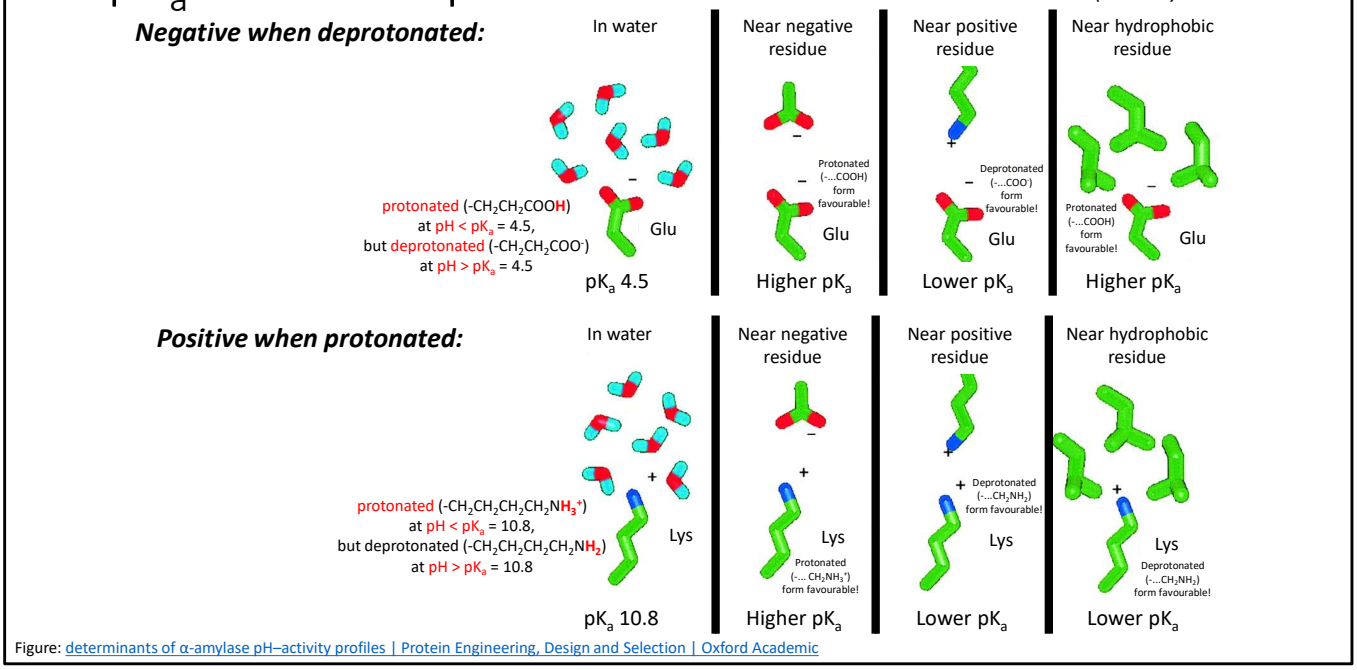


Figure: determinants of α-amylase pH–activity profiles | Protein Engineering, Design and Selection | Oxford Academic

The effect of placing a titratable group in a negative, positive and hydrophobic environment is summarized.

Let's look at an example. When the sidechain of glutamic acid is not in water, but inside a protein, in the vicinity of a positive charge of a basic residue such as lysine, a salt bridge between the negative sidechain of glutamic acid ($-CH_2CH_2COO^-$) and the positive sidechain of lysine ($-CH_2CH_2CH_2CH_2NH_3^+$) is possible. This situation favours the deprotonated, negatively charged state of the sidechain of glutamic acid. In this case, the pK_a value of the glutamic acid sidechain inside a protein in the vicinity of a positive charge is lower than the pK_a value of the glutamic acid sidechain in water: it becomes more acidic and "*gives up*" its proton more easily.

This is sometimes exploited in the active sites of enzymes to promote adapted reaction mechanisms that depend on the perturbed pK_a values of proton donor / acceptor groups of specific residues.

You Need to Be Able to...

- ❑ describe, recognize, and sketch common features of amino acids, such as their stereochemistry
- ❑ memorize, identify, use, and draw the names, abbreviations, structures, and properties of the 20 amino acids commonly found in proteins
- ❑ understand how the R groups of amino acids give them various properties
- ❑ list and spot common post-translational modifications of amino acids
- ❑ demonstrate the acid/base chemistry of amino acids and the meaning of pK_a
- ❑ predict how the pK_a of an amino acid may shift as a function of the environment
- ❑ construct titration curves of amino acids and calculate their isoelectric points using the easy-to-remember pK_a approximations

More Information?

- Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox
  - 1.2 Chemical Foundations: Three-Dimensional Structure Is Described by Configuration and Conformation
  - 2.2 Ionization of Water, Weak Acids, and Weak Bases
  - 2.3 Buffering against pH Changes in Biological Systems
  - 3.1 Amino Acids
- Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer
  - 1.3 Concepts from Chemistry Explain the Properties of Biological Molecules: Acid-base reactions are central in many biochemical processes; Acid-base reactions can disrupt the double helix; and Buffers regulate pH in organisms and in the laboratory
  - 2.2 Proteins Are Built from a Repertoire of 20 Amino Acids

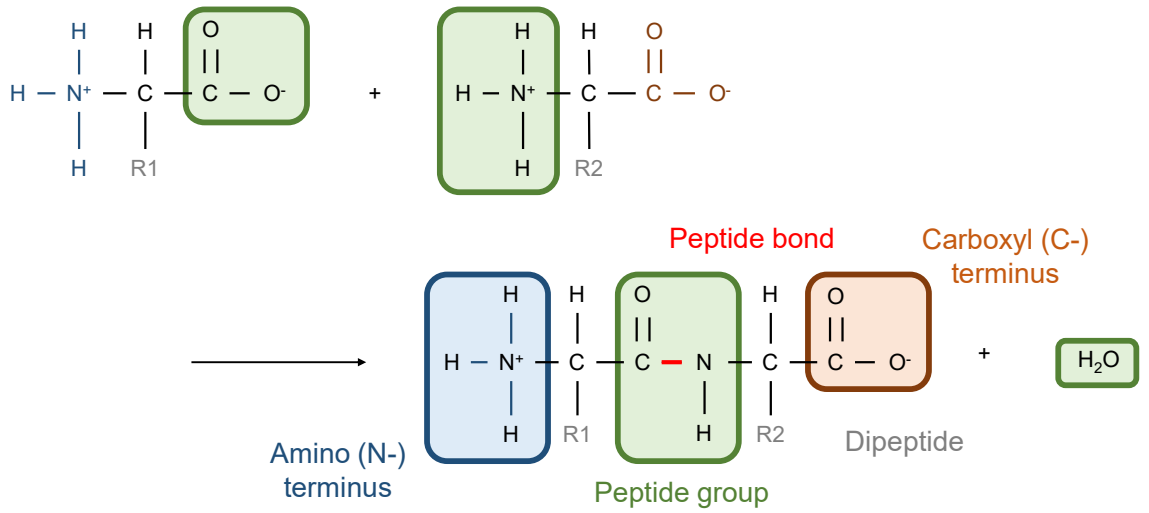
BIOC0001 – Proteins and Enzymes

Primary Structure

Dr. Renée Vancraenenbroeck
October 26, 2026

The **primary structure** of a protein is its linear sequence of amino acids. Here, we will look at properties of the **peptide bond** and the **naming** of peptides. We will also see how to calculate some **physical properties** of peptides: their molecular weight and isoelectric point (pI).

Formation of Peptide Bond



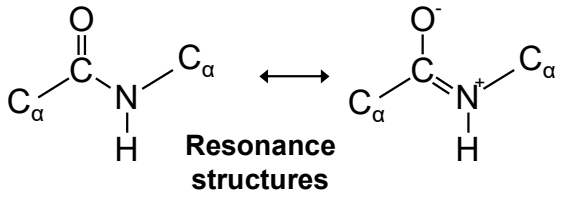
Two amino acid molecules can be covalently joined through an amide linkage to yield a **dipeptide**. The amide group (-CO-NH-) is referred to as the **peptide group**, the C-N bond in the peptide group is termed the **peptide bond**. Such an amide linkage is formed by removal of water from the α-carboxyl group ($-COO^-$) of one amino acid and the α-amino group ($-NH_3^+$) of another (*i.e.* a **condensation** reaction). This reaction requires energy.

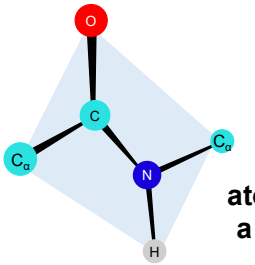
Amino acids that have been incorporated into proteins are called amino acid **residues**. The amino acid residue at the end with a free α-amino group is the **amino-terminal** (or *N*-terminal) residue; the residue at the other end, with a free α-carboxyl group, is the **carboxyl-terminal** (*C*-terminal) residue. By convention, peptides are drawn from left (the N-terminus) to right (the C-terminus). The C-terminus is available for formation of a peptide bond with another amino acid.

Eupeptide bonds are amide bonds formed between the α-carboxyl group of one amino acid and the α-amino group of another. **Isopeptide bonds** are amide bonds formed between a sidechain carboxyl group of one amino acid and the α-amino group of another or the α-carboxyl group of one amino acid and the sidechain amino group of another or …

P.S. Proteins absorb light at 190-220 nm due to the presence of the peptide bond.

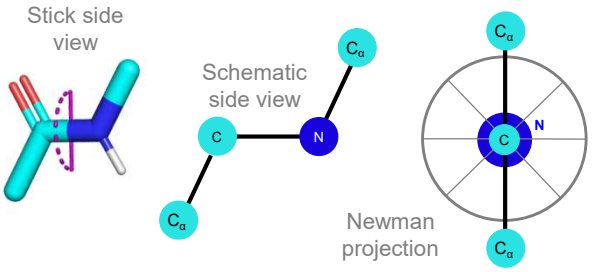
Planar, Rigid, and Trans



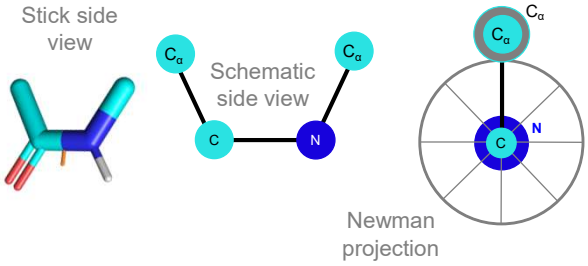


Six atoms in a plane

TRANS: C_α-C-N-C_α torsion angle, ω = 180°



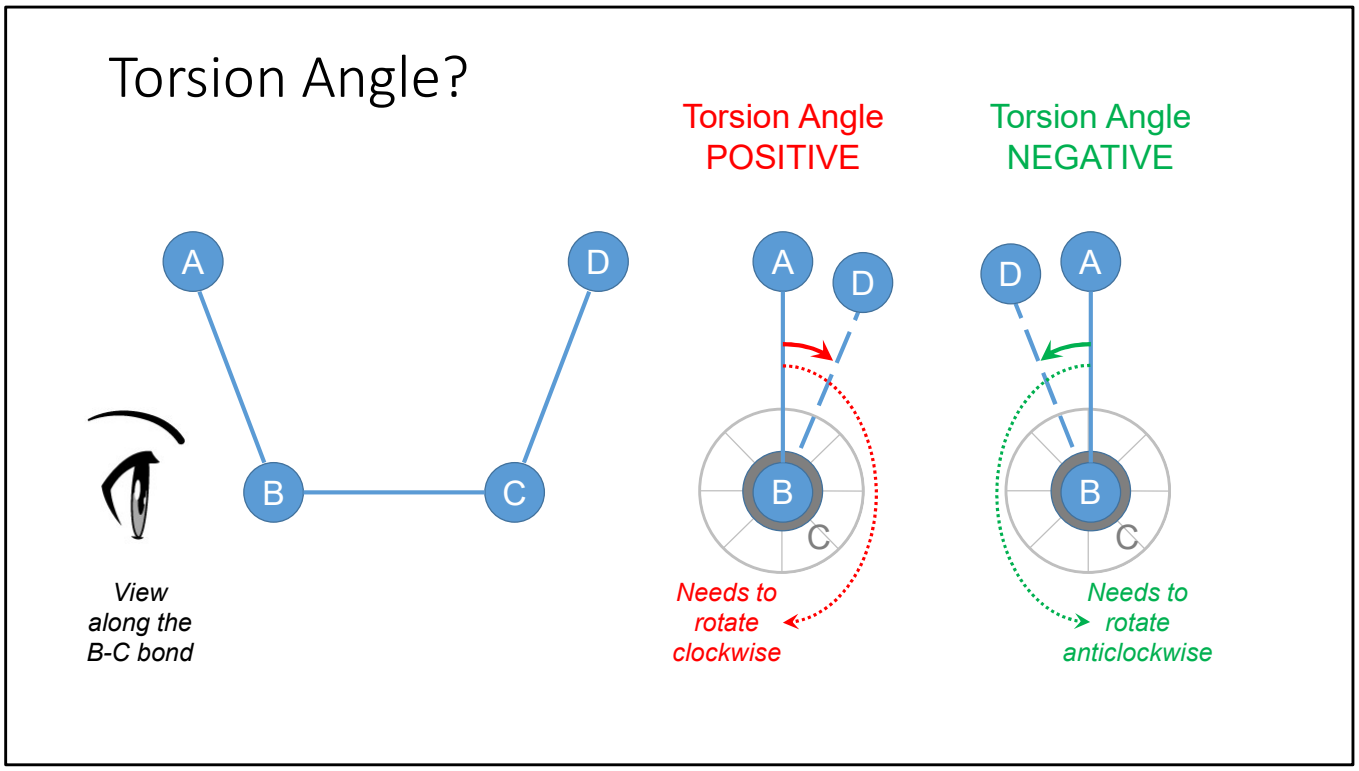
CIS: C_α-C-N-C_α torsion angle, ω = 0°



The peptide bond is **planar and rigid**. The result of **resonance** which gives rise to a **partial (40 %) double bond character**. Six atoms (C_α; C; O; N; H and C_α) lie in a plane. Rotation around the peptide bond is severely restricted.

Peptide bonds generally have the ***trans* configuration**. In the *trans* configuration, the two α-carbon atoms of the connected amino acids are on the opposite sides of the peptide bond. The **C_α-C-N-C_α torsion angle, ω, is 180°**. In the *cis* configuration the α-carbon atoms are on the same side of the peptide bond (ω = 0°). The *trans* preference can be explained by the steric clashes that occur between groups attached to the α-carbon atoms in *cis* form, which hinder formation of this configuration.

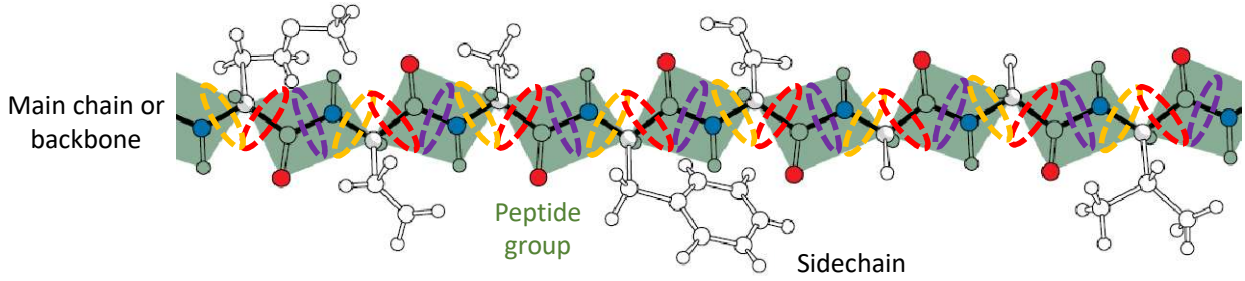
P.S. *Cis* configurations are sometimes found with proline residues.



A **torsion angle**, also called a **dihedral angle**, is defined by four sequentially bonded atoms. The left panel shows atoms A, B, C, and D illustrating the position of the four atoms used to define the dihedral angle. Imagine looking along the bond between atoms B and C, the middle bond of the three bonds defined (*i.e.* of bond A-B, bond B-C, and bond C-D).

The middle and right panels demonstrate the view along the B-C bond. This is the **Newman projection**. The torsion angle A-B-C-D is the apparent angle subtended by D relative to A, as shown. The torsion angle can have any value from -180° to +180°. If the chain is viewed along the B-C bond, the torsion angle is positive if the bond A-B needs to rotate in a clockwise sense (less than 180°) to align with the bond C-D (middle panel). Negative angles correspond to anticlockwise rotation (right panel).

Protein Backbone, ϕ, and ψ



Because of its partial double bond nature, the peptide bond (**omega, ω, C_α-C-N-C_α**) is always planar (0° (cis) or 180° (trans*, preferred*)).

Rotation is only possible around the **phi, ϕ, C-N-C_α-C** and **psi, ψ, N-C_α-C-N** bonds. The conformation of the backbone can therefore be described by ϕ and ψ torsion angles.

Figure: Voet & Voet's Biochemistry. 3rd edition. Figure copyrighted © by Irving Geis.

More help? Understanding PHI and PSI Angles, Ramachandran Plots, & Newman Projections

The linked series of carbon, nitrogen, and oxygen atoms in proteins are known as the **main chain or backbone (-NH-C_α-CO-)**.

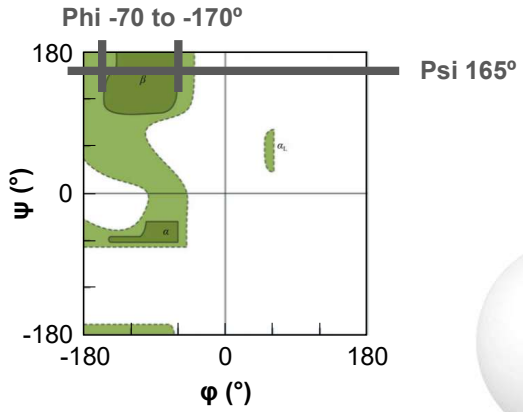
There are three main chain torsion angles of a peptide:

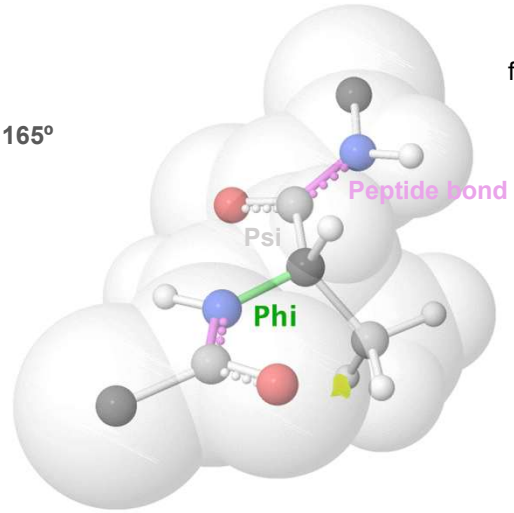
- **omega (ω: C_α-C-N-C_α)** = 180° due to the planar, rigid, and trans character of the peptide bond
- **phi (ϕ: C-N-C_α-C)** – in theory, all torsion angles are possible since N-C_α is a single bond
- **psi (ψ: N-C_α-C-N)** – in theory, all torsion angles are possible since C_α-C is a single bond

The nature of the peptide bond has implications for the detailed 3D structure of proteins! Since ω is fixed to 180°, only the ϕ and ψ torsion angles provide the flexibility required for the polypeptide backbone to adopt a certain fold. Thus, **rotation of the protein chain can be described as rotation of the peptide bond planes relative to each other**.

For more help, see the "Understanding PHI and PSI Angles, Ramachandran Plots, & Newman Projections" YouTube movie from Molecular Memory. Available through https://www.youtube.com/watch?v=JyUMLSsbecI.

Steric Hindrance and Ramachandran Plot





In the animation, Psi is fixed at +165º. Clashes occur at all Phi angles except ~ -70º to -170º. This range is in the β-strand region.

Figure: (IUCr) Half a century of Ramachandran plots
More help: Ramachandran Animation or Tutorial:Ramachandran principle and phi psi angles - Proteopedia, life in 3D

Although theoretically an infinite number of ϕ and ψ combinations can exist, only a limited number are possible due to **steric hindrance (collisions or clashes between atoms)**. Accessible values of the ϕ and ψ torsion angles are traditionally displayed in the form of **Ramachandran plots**, named after G.N. Ramachandran and co-workers who used computer models of small polypeptides to systematically vary ϕ and ψ torsion angles to find stable conformations.

The horizontal axis shows ϕ values, while the vertical shows ψ values. The white areas correspond to conformations where atoms in the polypeptide come closer than the sum of their **van der Waals radii**. These regions are **sterically disallowed**. The dark green regions correspond to conformations where there are no steric clashes, *i.e.* these are the **sterically allowed** regions *(we will see in later sessions that these correspond to the α-helical and β-sheet conformations)*. The light green areas show the allowed regions if slightly shorter van der Waals radii are used in the calculation, *i.e.* the atoms are allowed to come a little closer together.

The concept of steric hindrance is well demonstrated in the following interactive tutorial:
http://bioinformatics.org/molvis/phipsi/ or
https://proteopedia.org/w/Tutorial:Ramachandran_principle_and_phi_psi_angles
(same tutorial, different link – use the one that works in your browser).

Ramachandran Plot of Gly and Pro

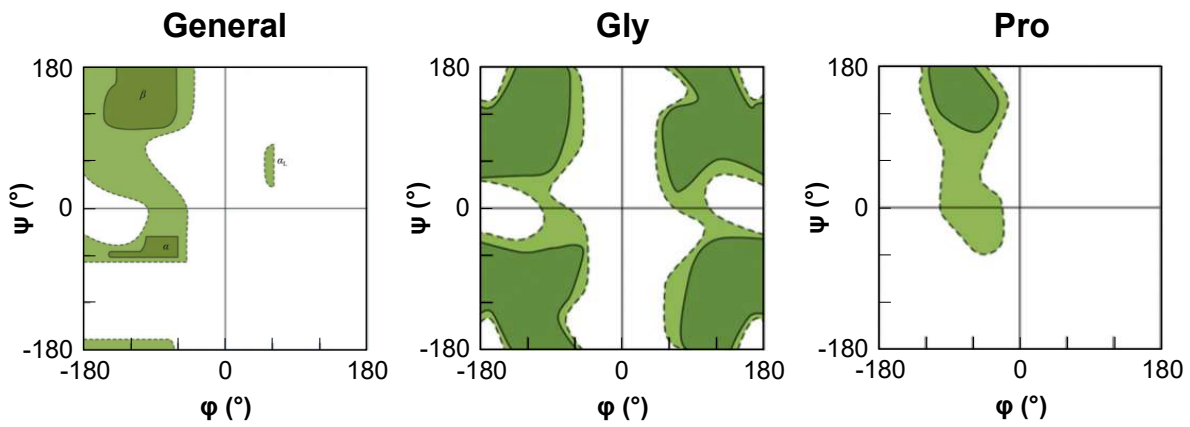


Figure: (IUCr) Half a century of Ramachandran plots

Glycine and proline are unique. Glycine has no side chain ("only" a hydrogen atom) and therefore can adopt ϕ and ψ angles in all four quadrants of the Ramachandran plot. Proline, on the other hand has a 5-membered ring as a side chain. Therefore, it is much more restricted than the other amino acids and allows for only a limited number of ϕ and ψ angles.

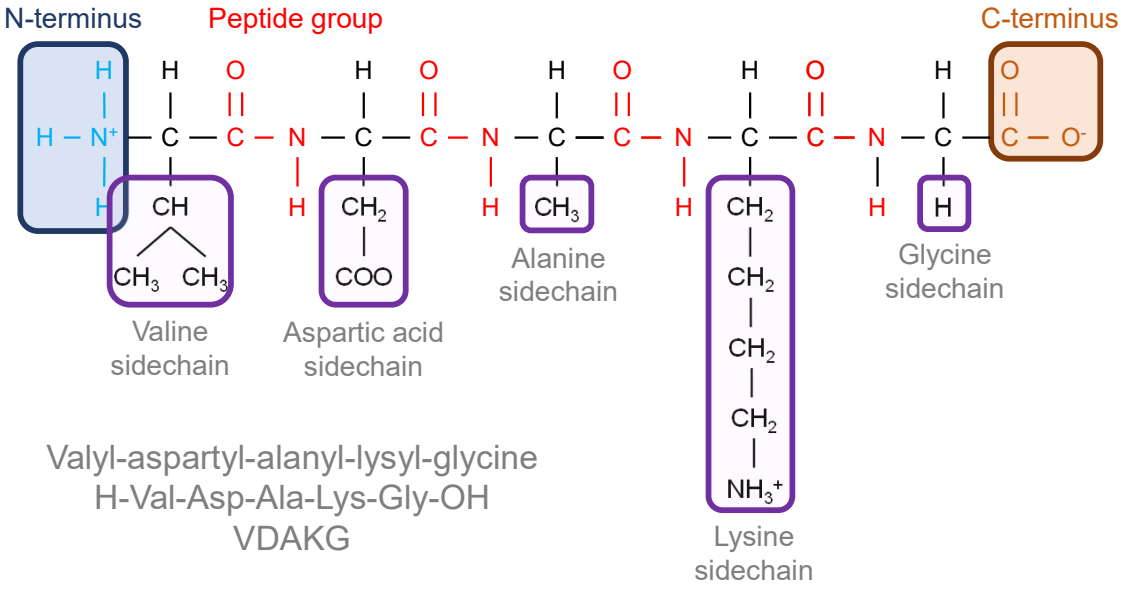
Oligopeptides, Polypeptides, and Proteins

	# amino acids	# peptide bonds
Dipeptide	2	1
Tripeptide	3	2
Tetrapeptide	4	3
…		
Oligopeptide	*< 25*	
Polypeptide	*< 100*	
Protein	*> 100, one or more polypeptide chains*	

Vague boundaries!

Two amino acid molecules can be covalently joined by one peptide bond to form a dipeptide. Three amino acids can be joined by two peptide bonds to form a tripeptide, *etcetera*.
When a few amino acids are joined, the structure is called an **oligopeptide**. When many amino acids are joined, the product is called a **polypeptide**. **Proteins** have a lot more amino acid residues and can exist out of one or more polypeptides. These are vague boundaries, and the terms “protein” and “polypeptide” are sometimes used interchangeably.

Naming

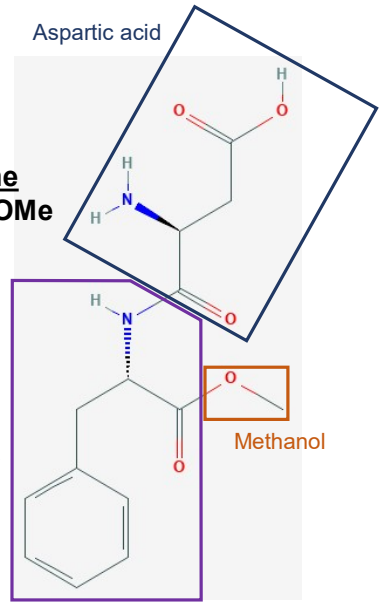


Names of peptides are derived from amino acid residues. By convention, names are written from left (the N-terminus) to right (the C-terminus). Full word names are very cumbersome for large peptides; we use the abbreviations.

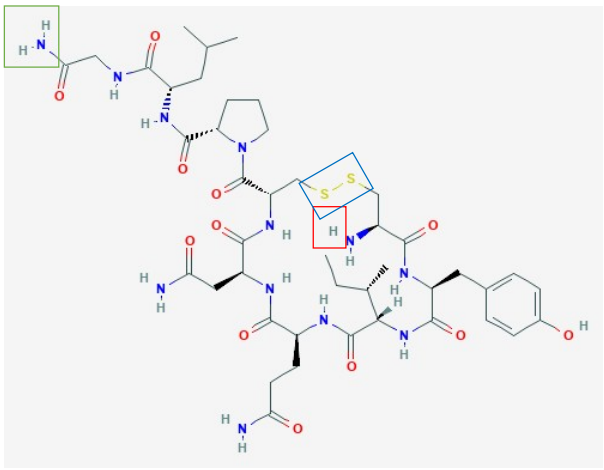
The pentapeptide Valyl-aspartyl-alanyl-lysyl-glycine, or H-Val-Asp-Ala-Lys-Gly-OH, or VDAKG is shown as an example.

Examples

Aspartame
H-Asp-Phe-OMe



Oxytocin
H-Cys(1)-Tyr-Ile-Gln-Asn-Cys(1)-Pro-Leu-Gly-NH2



The first six amino acid residues are drawn into a loop by a disulphide bond – represented with (1) behind both Cys. The H comes from the protonated NH, the NH2 refers to the glycinamide.

Figures from https://pubchem.ncbi.nlm.nih.gov/compound/Aspartame and https://pubchem.ncbi.nlm.nih.gov/compound/Oxytocin

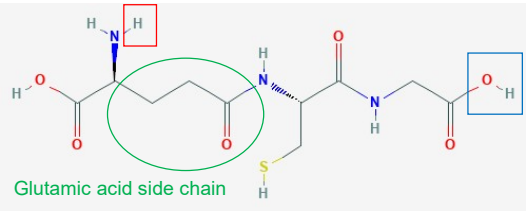
Aspartame is the name for an artificial, non-carbohydrate sweetener, aspartyl-phenylalanine methyl ester, *i.e.* the methyl ester of the dipeptide of the amino acids aspartic acid and phenylalanine. It is marketed under several trademark names. It is commonly used in diet soft drinks and is often provided as a table condiment.

Oxytocin (cysteinyl-tyrosyl-isoleucyl-glutaminyl-asparagyl-cysteinyl-prolyl-leucyl-glycinamide (1->6)-disulphide) is produced by the pituitary gland and stimulates uterine contractions in labour. Oxytocin was the first naturally occurring hormone to be produced in a laboratory. It is a cyclic nonapeptide hormone with amino acid sequence CYIQNCPLG. The first six amino acid residues are drawn into a loop by a disulphide bond.

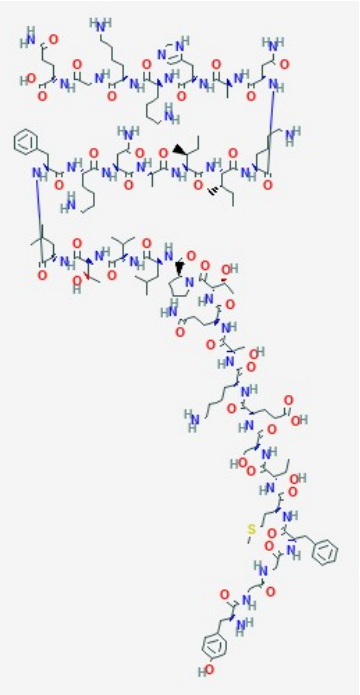
Examples *(cont.)*

Glutathione

H-γGlu-Cys-Gly-OH



The γ-carboxyl group of the glutamic acid residue (not the usual α-carboxyl group!) contributes to the peptide bond. This is an example of an isopeptide bond.



β-endorphin

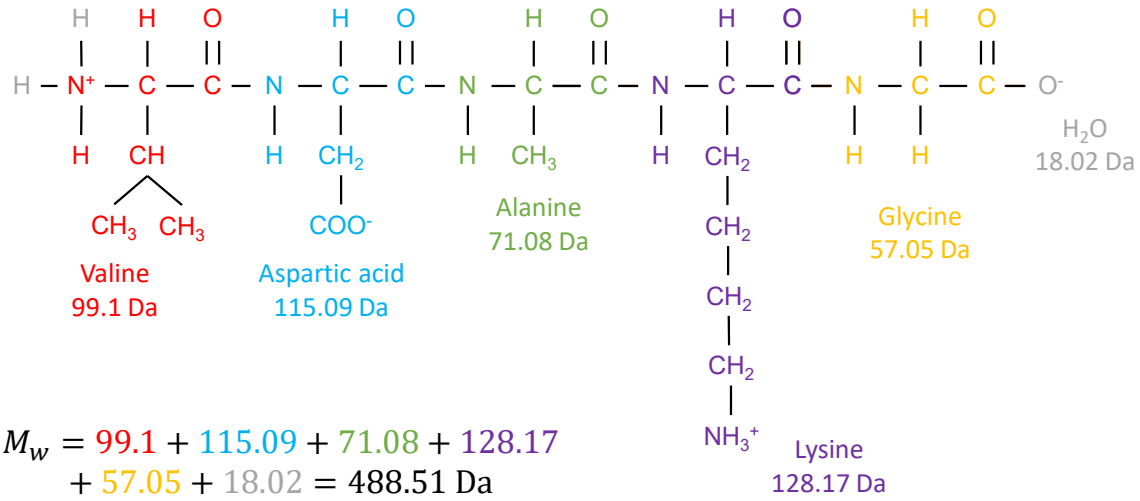
H-Tyr-Gly-Gly-Phe-Met-Thr-Ser-Glu-Lys-Ser-Gln-Thr-Pro-Leu-Val-Thr-Leu-Phe-Lys-Asn-Ala-Ile-Ile-Lys-Asn-Ala-His-Lys-Lys-Gly-Gln-OH

Figures from https://pubchem.ncbi.nlm.nih.gov/compound/Glutathione and https://pubchem.ncbi.nlm.nih.gov/compound/beta-ENDORPHIN

Glutathione (γ-glutamyl-cysteinyl-glycine) is a tripeptide where the γ-carboxyl group of the glutamic acid residue (not the usual α-carboxyl group!) contributes to the peptide bond. It is a very effective reducing agent and is involved in protecting cells from the destructive effects of oxidation. It is also important as a cofactor for the enzyme glutathione peroxidase.

Endorphins are natural painkillers that are predominantly produced in the pituitary gland and the brain. They interact with receptors in the brain to inhibit the transmission of pain signals. An example of an endorphin peptide is **β-endorphin** (tyrosyl-glycyl-glycyl-phenylalanyl-methionyl-threonyl-seryl-glutamyl-lysyl-seryl-glutaminyl-threonyl-prolyl-leucyl-valyl-threonyl-leucyl-phenylalanyl-lysyl-asparagyl-alanyl-isoleucyl-isoleucyl-lysyl-asparagyl-alanyl-histidyl-lysyl-lysyl-glycyl-glutamine). It contains 31 amino acid residues and is the C-terminal fragment of β-lipotropin.

Physical Properties: M_w

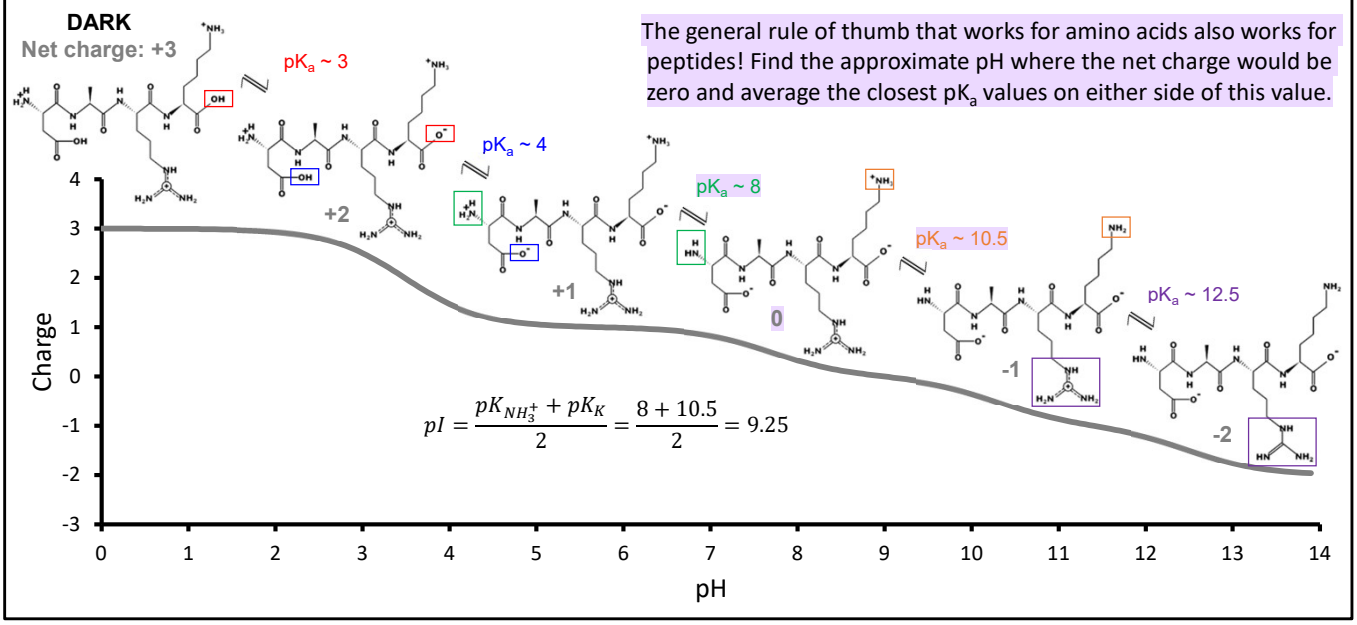


$$M_w = 99.1 + 115.09 + 71.08 + 128.17 + 57.05 + 18.02 = 488.51 \text{ Da}$$

Peptide **molecular weight** or **M_w** is calculated by the addition of the isotopic masses of amino acids (those already reflect the loss of water during bond formation) in the peptide and the isotopic mass of one water molecule. The molecular weight is given in **Dalton, 1 Da = 1 g/mol**.

The average isotopic mass of an amino acid residue in a peptide is 110 Da. Comes in handy when you want to approximate the molecular weight of a protein!

Physical Properties: pI



The characteristic pH at which the net charge is 0 is called the **isoelectric point** or **pI**. For peptides, the isoelectric point can be determined by calculating the average of the two pK_a values which are closest to the pH at which the net charge of the protein is zero.

You Need to Be Able to...

- ❑ describe and draw the structure and properties of the peptide bond
- ❑ understand how the shape of polypeptides is influenced and limited by the rotation around various bonds in the peptide
- ❑ identify and measure ω, ϕ and ψ torsion angles
- ❑ explain ϕ and ψ torsion angles within the context of steric hindrance and the Ramachandran plot
- ❑ know how to name an oligopeptide
- ❑ list examples of oligopeptides (number of amino acids, origin, and role)
- ❑ discuss what kind of information can be obtained from primary structure
- ❑ calculate the isoelectric point and molecular weight of a peptide

More Information?

- Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox
  - 3.2 Peptides and Proteins
  - 3.4 The Structure of Proteins: Primary Structure (but not “Protein Chemistry”, “Mass Spectrometry” and “Chemical Synthesis”)
  - 4.1. Overview of Protein Structure
- Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer
  - 2.3 Primary Structure: Amino Acids Are Linked by Peptide Bonds to Form Polypeptide Chains
  - 10.1 Homologs Are Descended from a Common Ancestor and Can Be Detected by Sequence Alignments
  - 10.3 Evolutionary Trees Can Be Constructed on the Basis of Sequence Information

BIOC0001 – Proteins and Enzymes

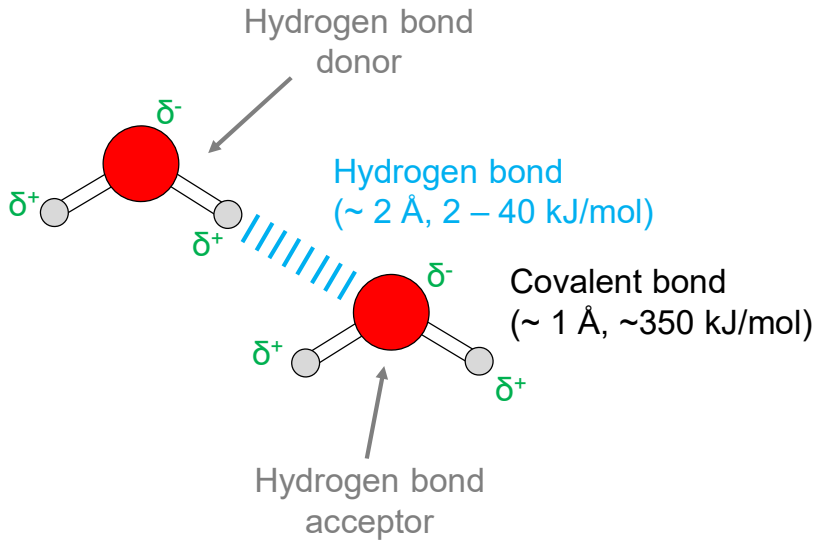
Secondary Structure

Dr. Renée Vancraenenbroeck
October 26, 2026

Here, we will explore protein **secondary structure**.

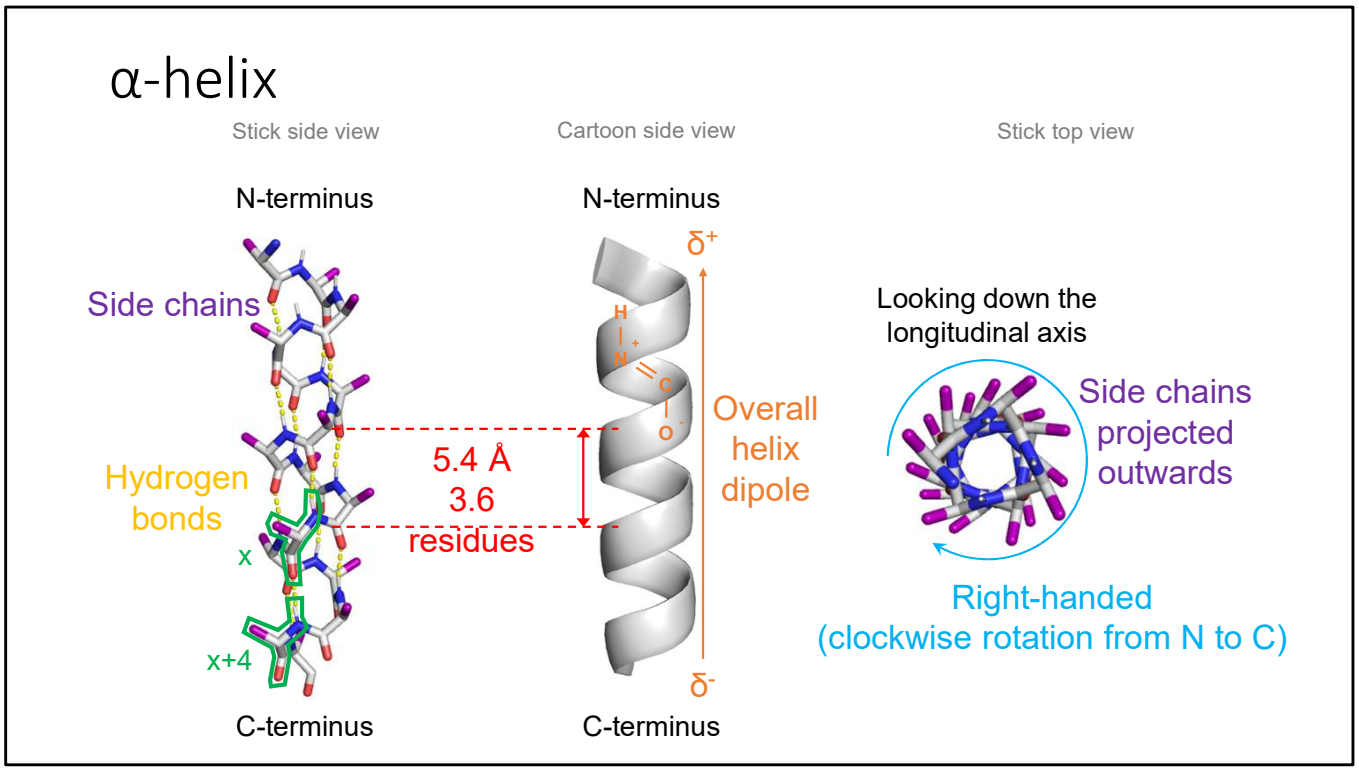
Secondary structure is the local spatial arrangement of the polypeptide's backbone atoms ignoring the conformations of its side chains. It includes the stable and widely occurring **α-helices, β-sheets, and β-turns**.

Hydrogen Bonds



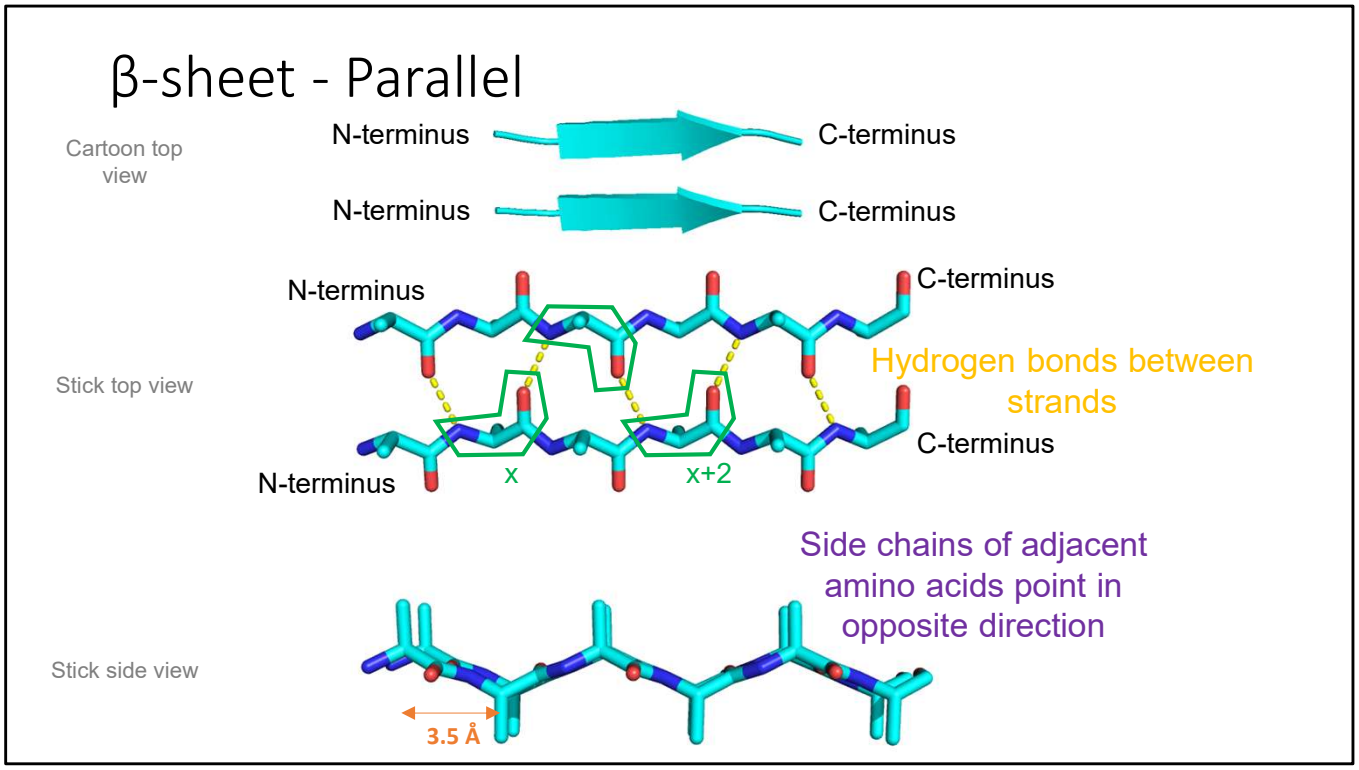
Hydrogen bonds are crucial to α-helices and β-sheets. Hydrogen atoms on electronegative atoms, *e.g.* **N-H** or **O-H**, are sufficiently electron poor that they will interact with lone pairs usually on **N** or **O** atoms. This is because the bonding pair is pulled towards the nitrogen or oxygen atom, which leaves that end of the bond partially negative (δ^-) and the hydrogen end partially positive (δ^+).
The group that provides the hydrogen atom is referred to as the **hydrogen bond donor**; the group that provides the lone pair is referred to as the **hydrogen bond acceptor**.

Hydrogen bonds are much weaker than covalent bonds, yet they are strong enough - when many form simultaneously - to help stabilize specific structures! They have a bond dissociation energy (*i.e.* the energy required to break the bond) between **2 to 40 kJ/mol**, compared to ~350 kJ/mol for the covalent C-C bond. Hydrogen bonds are also longer than covalent bonds; their bond distances range from **~ 2 Å**, compared to ~ 1 Å for a single covalent bond. The strongest hydrogen bonds have the hydrogen-bond donor, the hydrogen atom, and the hydrogen-bond acceptor along a **straight** line.



The **α-helix** was first described by Linus Pauling, Robert Corey, and Herman Branson.

- It is stabilised by **hydrogen bonds between C=O and N-H groups that are four residues along, *i.e.* x and x+4**. Except for amino acids near the ends of an α-helix, all the main-chain C=O and N-H groups are hydrogen bonded.
- The **side-chains project outwards** from the α-helix; thus, steric interference with the backbone or with other sidechains is avoided.
- Each residue is related to the next one by a rise of 1.5 Å along the axis and a rotation of 100 degrees, which gives **3.6 amino acid residues per turn**. Thus, amino acids spaced three and four apart in the sequence, *i.e.* x and x+3 and x+4, are spatially quite close to one another in an α-helix. The **pitch** of the α-helix, which is the distance for one complete turn, is equal to the product of the rise between two amino acids (*i.e.* 1.5 Å) and the number of residues per turn (*i.e.* 3.6), **is 5.4 Å**.
- The screw sense of an α-helix can be **right-handed** (rotation is clockwise as the polypeptide chain is followed N to C) or left-handed (counterclockwise). Yet essentially all α-helices found in proteins are right-handed.
- A small electric dipole exists in each peptide bond. These dipoles are connected through the hydrogen bonds of the helix, resulting in an overall **helix dipole**, extending along the helix that increases with helix length.



Another common secondary structure is the **β-sheet**, which contains extended stretches of polypeptide chain with **hydrogen bonds between neighbouring strands** between C=O and N-H groups. In a **parallel** β-sheet, polypeptide strands run in the same direction (*i.e.* from N- to C-terminus), whereas in an **antiparallel** β-sheet, neighbouring strands extend in opposite directions.

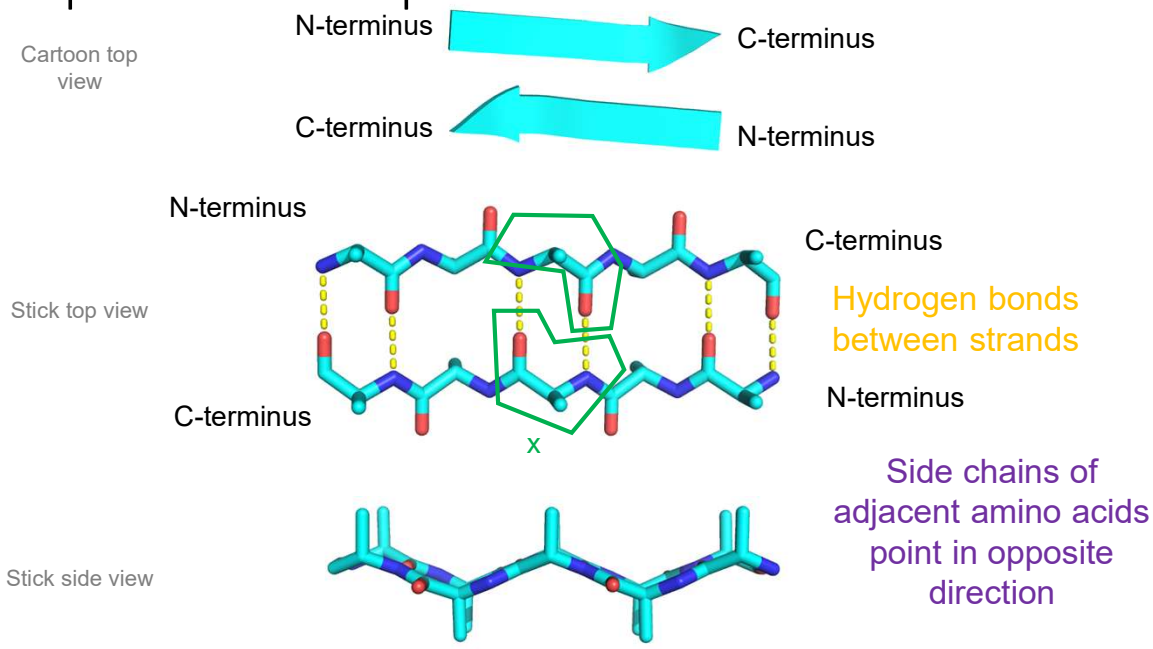
In the parallel arrangement, for each amino acid, the N-H group is hydrogen bonded to the C=O group of one amino acid on the adjacent strand, whereas the C=O group is hydrogen bonded to the N-H group on the amino acid two residues along the chain.

The strand in a β-sheet is almost fully extended. The distance between adjacent amino acids along a strand is approximately 3.5 Å (in contrast with a distance of 1.5 Å along an α-helix).

The **side chains of adjacent amino acids point in opposite direction**.

Many strands, typically 4 or 5, but more are possible, can come together in β-sheets. The β-sheets can be purely antiparallel, purely parallel, or mixed.

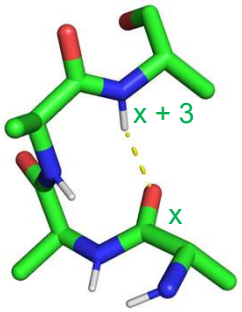
β-sheet - Antiparallel



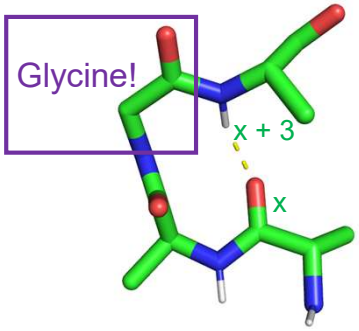
In the antiparallel arrangement, the N-H group and the C=O group of each amino acid are respectively hydrogen bonded to the C=O group and the N-H group of a partner on the adjacent chain.

Loops and Turns

β-turn Type I



β-turnType II



Type	ϕ_{x+1}	ψ_{x+1}	ϕ_{x+2}	ψ_{x+2}
I	−60°	−30°	−90°	0°
II	−60°	120°	80°	0°

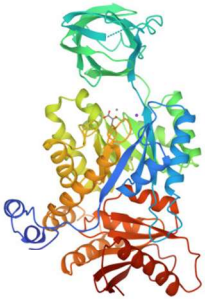
Loops and **turns** connect α-helices and β-sheets and can cause a change in direction of the polypeptide's chain. Loops can vary in lengths and shapes. They are generally located on the surface of the protein and are rich in polar amino acids. In general, α-helices and β-sheets are evolutionary more conserved than loops. Small loops with only 3 to 6 amino acids, a regular hydrogen bond pattern, and specific dihedral angles are called turns. There are many classes of turns, the most common one is the β-turn.

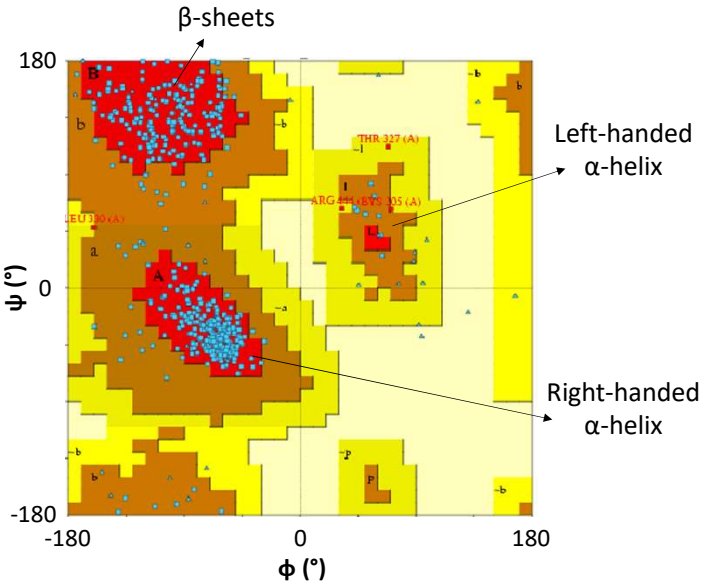
A **β-turn** (also known as a reverse turn) is a 180° turn involving **four amino acid residues**. The **C=O group of residue *x* is hydrogen bonded to the N-H group of residue *x* + 3**. This interaction stabilizes abrupt changes in direction of the polypeptide chain.
β-turns are **further divided into classes based on the ϕ and ψ angles of the residues at positions *x* + 1 and *x* + 2**. Types I and II are the most common. Type I turns have approximately ϕ_{x+1} = -60°, ψ_{x+1} = -30°, ϕ_{x+2} = -90°, ψ_{x+2} = 0°, and type II approximately ϕ_2 = -60°, ψ_2 = 120°, ϕ_3 = 80°, ψ_3 = 0°. The ***x* + 2 residue of the type II turn** can only be occupied by **glycine**.

Ramachandran Plot

RABBIT MUSCLE PYRUVATE KINASE COMPLEXED WITH MN^{2+}, K^{+}, AND PYRUVATE

1PKN





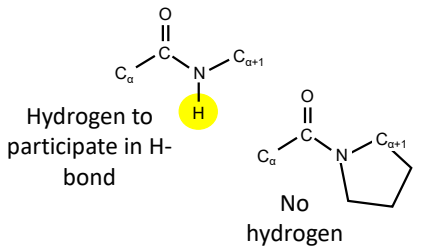
Figures: RCSB PDB - 1PKN: STRUCTURE OF RABBIT MUSCLE PYRUVATE KINASE COMPLEXED WITH MN2+, K+, AND PYRUVATE and PROCHECK summary for 1pkn

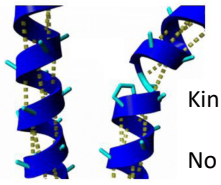
More help? Understanding PHI and PSI Angles, Ramachandran Plots, & Newman Projections

All secondary structure elements fall within the 'allowed' conformation areas of the Ramachandran plot. The different secondary structures can be distinguished by their range of ϕ and ψ values with the values of different secondary structures mapping to different regions of the Ramachandran plot.

For more help, see the "Understanding PHI and PSI Angles, Ramachandran Plots, & Newman Projections" YouTube movie from Molecular Memory. Available through https://www.youtube.com/watch?v=JyUMLSsbecI.

Propensity Scores





Kink in helix

No amide bound hydrogen

	P_α	P_β	P_{turn}
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.47
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56
Asn	0.67	0.89	1.56

Certain amino acids are observed more frequently than others in α-helices, β-sheets or β-turns, reflecting the properties of the R-group and the ϕ and ψ angles it can adopt. Chou and Fasman investigated this phenomenon in 1974. They calculated the **propensity** of each amino acid to participate in α-helices, β-sheets or β-turns, *i.e.* P(α), P(β), and P(turn), based on observed frequencies in a set of sample proteins of known structure. They concluded that:

- Ala, Met, and Glu are α-helix formers,
- Tyr, Ile, and Val are β-sheet formers,
- Asn, Gly, and Pro are most frequent in β-turns.

Proline is a peculiar amino acid. Prolines are rarely found in α-helices and β-sheets:

- The nitrogen atom of a proline residue in a protein has no substituent hydrogen to participate in hydrogen bonds with other residues further down the chain.
- In proline, rotation about the N-C_α bond is not possible. Proline introduces a destabilizing kink in an α-helix which often disrupts helix formation.

Obviously, the position of an amino acid residue relative to its **neighbours** is also important. For example, if a polypeptide chain has a long block of Glu residues, this segment of the chain will not form an α-helix at pH 7.0. The negatively charged carboxyl groups of adjacent Glu residues repel each other so strongly that they prevent formation of the α-helix.

You Need to Be Able to...

- ❑ illustrate the role of hydrogen bonds between main-chain atoms in secondary structure elements
- ❑ describe and analyse the structure and properties of α-helices, β-sheets, and β-turns
- ❑ recognize how secondary structure is defined by the permitted ϕ and ψ angles of the amino acids within the segment and how the Ramachandran plot visually depicts this
- ❑ explain and apply the conformational preferences of amino acids within secondary structure elements

More Information?

- Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox
  - 2.1 Weak Interactions in Aqueous Systems (but not "Osmotic Pressure")
  - 4.2 Protein Secondary Structure (but not "Circular Dichroism")
- Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer
  - 1.3 Concepts from Chemistry Explain the Properties of Biological Molecules: Covalent bonds and noncovalent interactions are important for structure and stability of biological molecules; and The formation of DNA's double helix is an expression of the rules of chemistry
  - 2.4 Secondary Structure: Polypeptide Chains Can Fold into Regular Structures

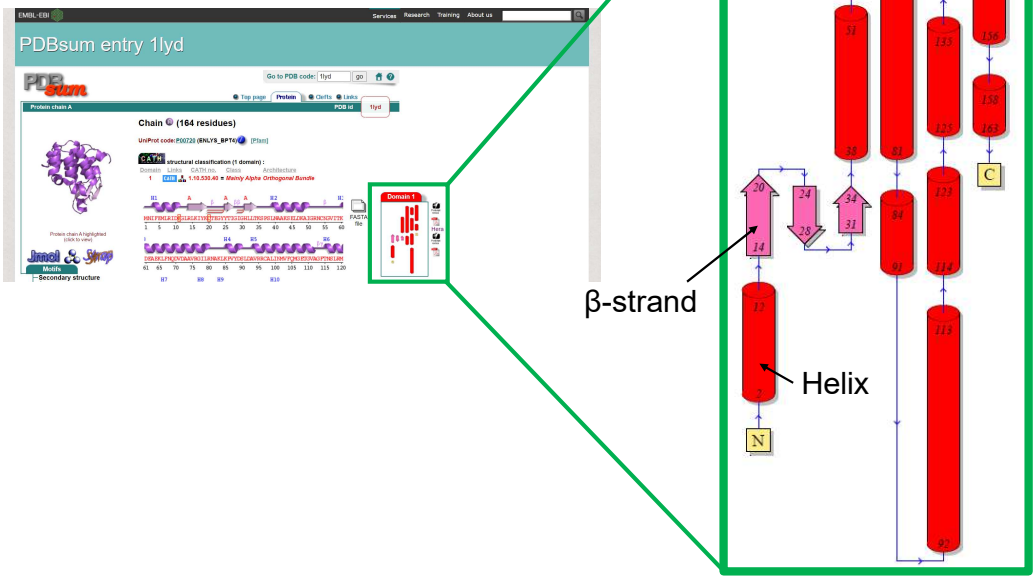
BIOC0001 – Proteins and Enzymes

Tertiary and Quaternary Structure

Dr. Renée Vancraenenbroeck
October 26, 2026

Here, we will discuss protein **tertiary and quaternary structure**. In addition, we will investigate **receptor-ligand binding interactions** and **protein folding and denaturation**.

Topology Diagrams

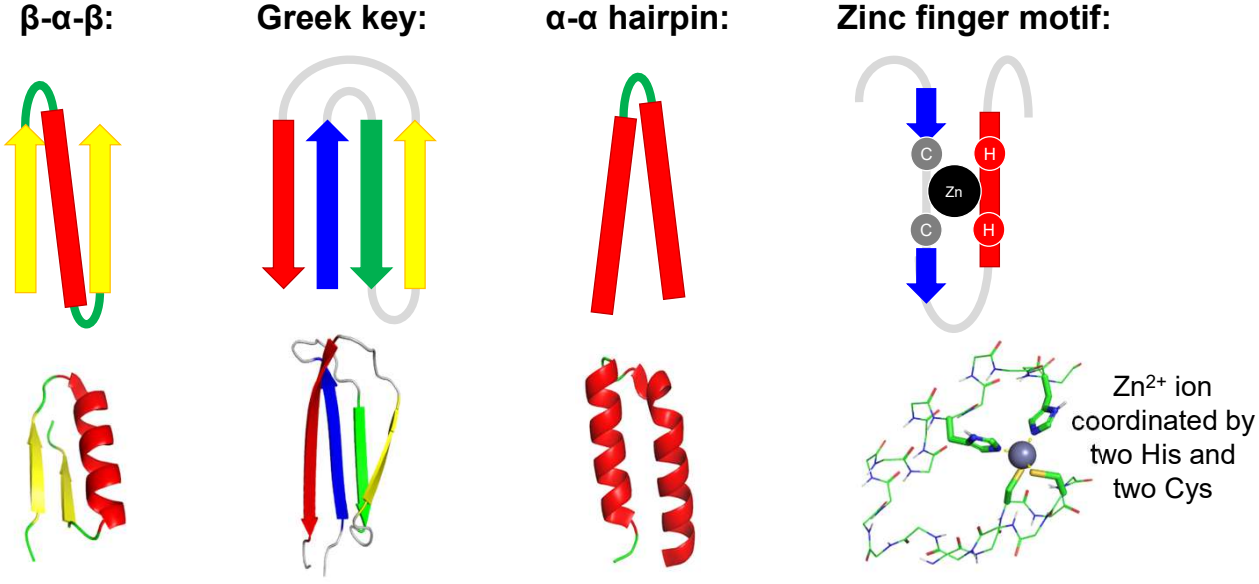


Figures: PDBsum home page

We have already heard of the **Protein Data Bank (PDB)**: https://www.rcsb.org/. Also of interest is the **PDBSum** database: http://www.ebi.ac.uk/pdbsum, a pictorial database that provides an at-a-glance overview of every macromolecular structure deposited in the PDB. It provides schematic diagrams of the molecules in each structure and of the interactions between them.

The PDBSum also includes **topology diagrams** which are useful for visualising the arrangement and connectivity of secondary structure elements of proteins. They provide a highly simplified description of the protein fold including the sequence of secondary structure elements, their relative spatial positions, and their approximate orientations. β-sheets are represented by arrows, α-helices by cylinders (or rectangles).

Super-secondary Structures



PDB: 1DBW, FixJ receiver domain; 1TTG, fibronectin; and 1EI0, P8MTCP1, PDB: 1ZNF, Zinc finger from Xfin protein;

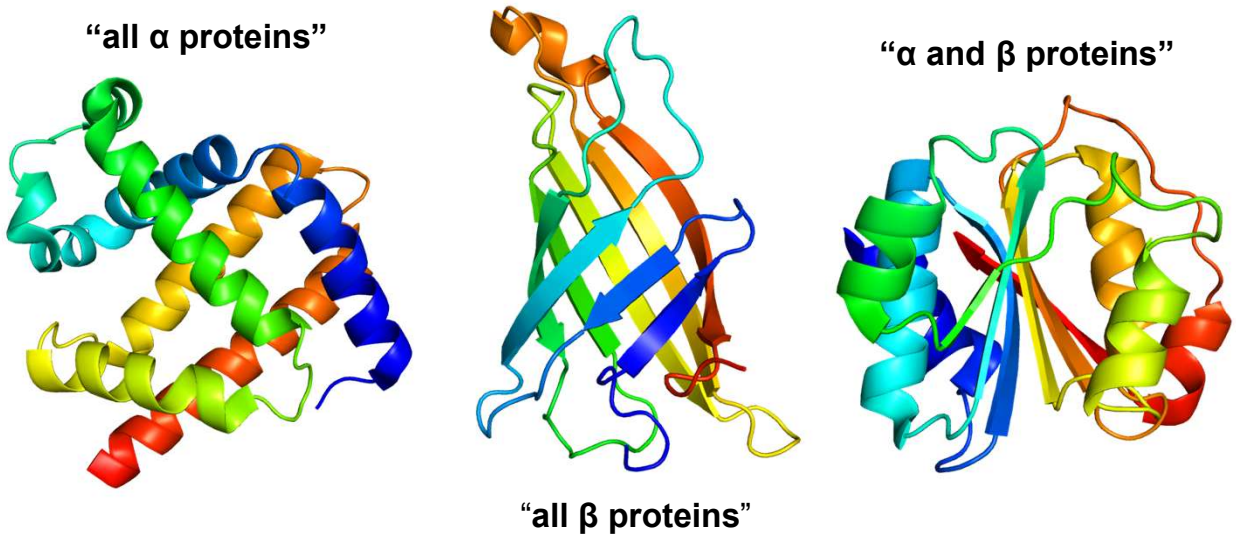
Within protein folds, **super-secondary structures or motifs**, *i.e.* distinctive arrangements of elements of secondary structure, have been described. This level of organisation is intermediate between secondary and tertiary structure. These motifs occur in many different proteins, with quite distinct amino acid sequences. Nonetheless, the chemical nature of the component residues (charged, polar or non-polar) is critical in many cases.

Examples of super-secondary structures include:

- **β-α-β motifs**: Antiparallel β-strands can be linked by short turns. In contrast, parallel β-strands are connected by longer regions of chain which cross the β-sheet and frequently contain α-helical segments. Hydrophobic residues on the α-helix and β sheets interact.
- **Greek key**: The Greek key topology is an example of a larger motif. It is named after a pattern that was common on Greek pottery. It is formed by three consecutive antiparallel beta-strands connected by hairpins followed by a longer connection to the fourth strand, which lies adjacent to the first; producing a β-sandwich.
- **α-α hairpins**: A helix hairpin or α-α hairpin refers to the loop connecting two antiparallel α-helical segments. The longer the length of the loop, the greater the number of possible conformations. For loops of two or three residues, there is only one allowed conformation. Antiparallel α-helices will interact generally by hydrophobic interactions between side chains at the interface.

- **Zinc fingers***:* This finger-like structure – often in nucleic acid–binding proteins – consists of an α-helix and two short antiparallel β-strands all held together by a **Zn^{2+}** ion. The Zn^{2+} ions are tetrahedrally coordinated by the side chains of Cys, His, and occasionally Asp or Glu. Zinc fingers are too small to be stable in the absence of Zn^{2+}. This is a nice example of how compounds like **metal ions and ligands** may stabilize tertiary structure!

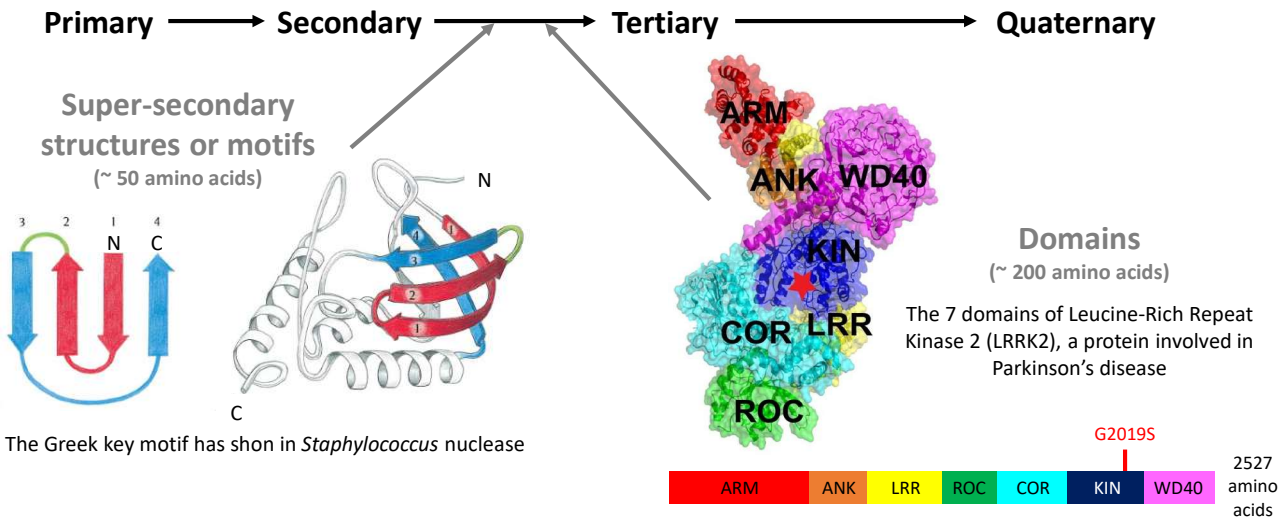
Tertiary Structure



PDB: 3RGK, myoglobin; 4I60, avidin; and 6ADH, horse liver alcohol dehydrogenase

Tertiary structures of proteins can be described by the specific topological arrangement of the secondary or super-secondary structural motifs. Some proteins consist only of α-helices spanned by short connecting links – called “**all α proteins**”, others have a large proportion of β-sheets and are basically devoid of α-helices – called “**all β proteins**”. Most proteins have significant amounts of both types of secondary structure – called “**α and β proteins**”.

Domains

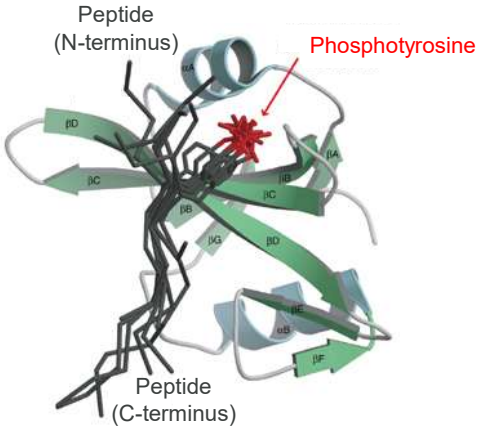


Figures: Branden & Tooze, Introduction to protein structure and Closing the structure-to-function gap for LRRK2 - ScienceDirect

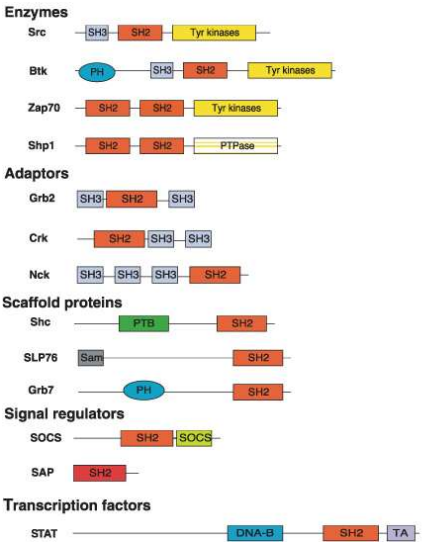
Proteins containing more than ~200 residues usually fold into two or more globular clusters known as **domains**. These domains often have a specific function, *e.g.* ligand binding. Many domains are structurally independent units that have the characteristics of small globular proteins. They are stable and often fold independently of the rest of the protein chain.

For example, LRRK2 has 7 domains. Two are enzyme domains: a GTPase (ROC) and a kinase (KIN).

Domains *(cont.)*



Eight different SH2/phosphopeptide complex structures were used to show **different modes of peptide recognition by the SH2 domain** .The phosphotyrosine residue is shown in red.



Families of SH2 domain containing proteins. Examples include enzymes, adaptors, scaffold proteins, signal regulators, and transcription factors.

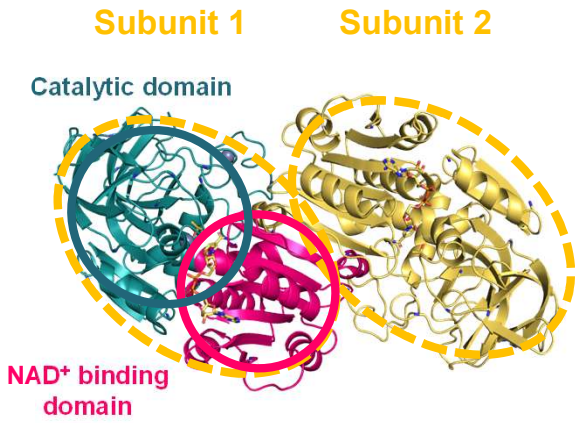
Figure: SH2 and PTB Domains in Tyrosine Kinase Signaling | Science's STKE

Another example is the **SH2 phosphotyrosine binding domain**. SH2 domains bind peptide motifs containing phosphorylated tyrosine residues. Structurally, SH2 domains tend to be highly conserved, comprising a central antiparallel β-sheet flanked by two α-helices.

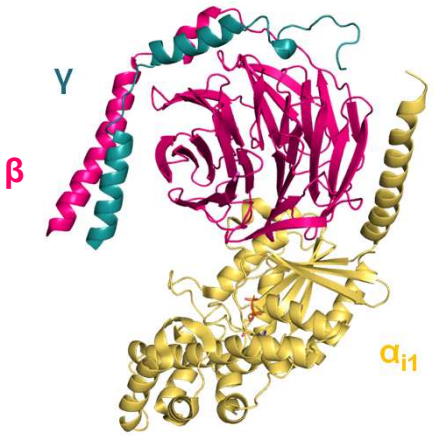
Approximately 100 proteins containing SH2 domains have been identified. SH2 domains are typically found in proteins involved in phosphotyrosine dependent signal transduction and serve as modulators of protein-protein-interactions.

Quaternary Structure

HOMO-DIMER



HETERO-TRIMER



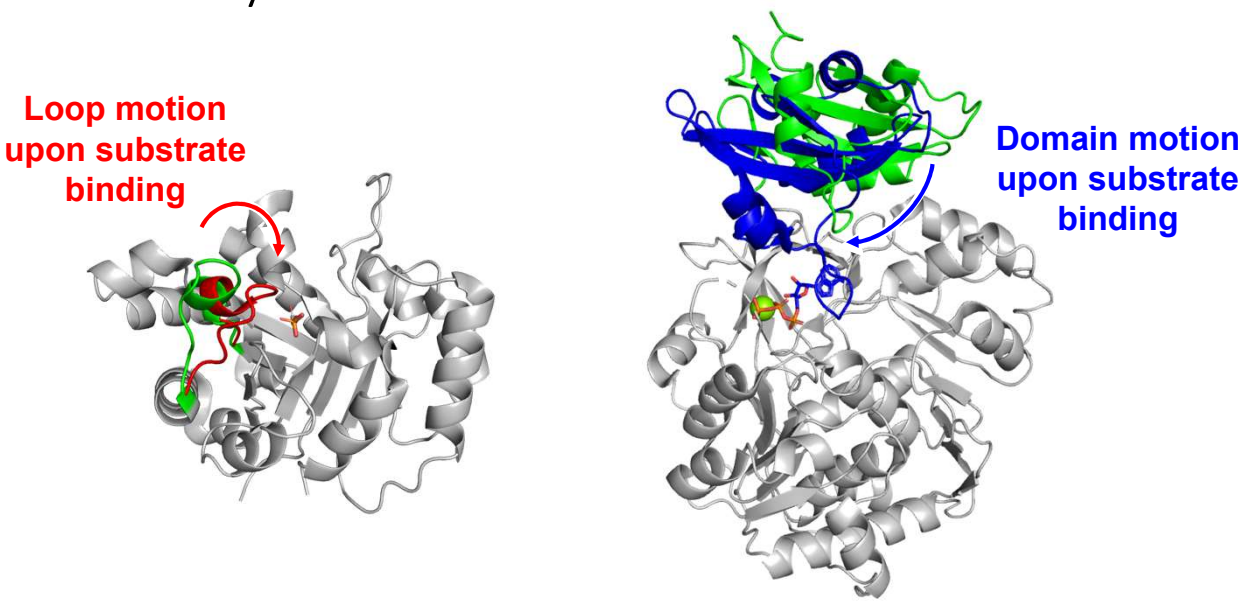
PDB: 6ADH, horse liver alcohol dehydrogenase and 1GP2, G protein hetero-trimer

Quaternary structure describes a protein composed of two or more polypeptides, called **subunits**. A *dimer* contains two, a *trimer* three, a *tetramer* four subunits... Multimers made up of identical subunits are referred to with a prefix of **"homo-"** (*e.g.* a homo-dimer); multimers made up of different subunits are referred to with a prefix of **"hetero-"** (*e.g.* a hetero-trimer).

For example, alcohol dehydrogenase is a homodimer. Each subunit has 374 residues with molecular weight of 74 kDa. There are two domains per subunit: the NAD^+-binding domain (residues 176-318) and the catalytic domain (residues 1-175, 319-374), both have an α/β structure. The G protein heterotrimer $\alpha_{i1}\beta_1\gamma_2$ has an α subunit which binds GDP with high affinity. The binding is stabilized by subunits β and γ.

Like tertiary structure, multimeric polypeptides are formed by the same kinds of non-covalent interactions and may be stabilized by disulphide bridges.

Flexibility

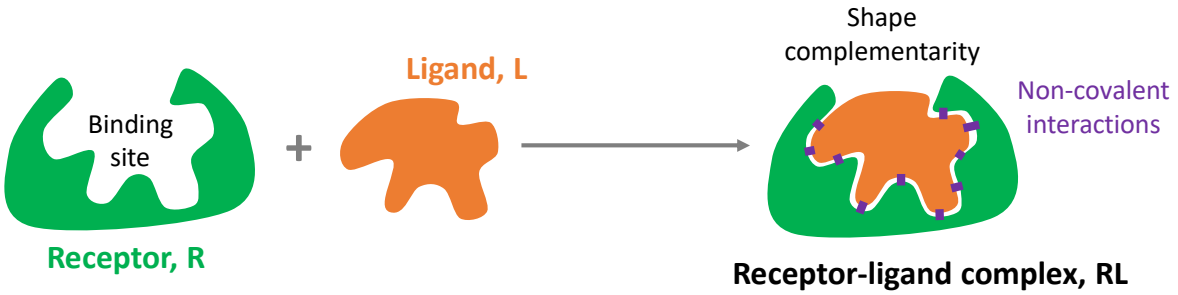


PDB: 1TPD and 1TRD, trypanosomal triosephosphate isomerase (TIM) and 4FUQ and 4FUT, *Rhodopseudomonas palustris*, malonyl-CoA synthetase

Loop motions are often key to protein function. For example, when the substrate binds the triosephosphate isomerase enzyme, a loop changes from an 'open' conformation (green) to a 'closed' one (red), which prevents solvent access to the active site and stabilises the intermediate compound of the reaction.

But also **domain motions** can be important in protein function. For example, bacterial malonyl-coenzyme A (CoA) synthetase undergoes a conformational change when the substrate binds: the C-terminal lobe rotates upon ATP / Mg^{2+} binding.

Receptor-ligand Interactions



Receptor-ligand interactions are among the most fundamental functions in biochemistry. The receptor is generally a macromolecule (protein or nucleic acid) and the ligand is a smaller molecule or another macromolecule. Receptor and ligand associate via **non-covalent interactions**, such as hydrogen bonding, electrostatic interactions, hydrophobic and Van der Waals forces, and **shape complementarity**. The **binding site** is the site on the receptor that binds to the ligand.

Examples of receptor-ligand interactions and their physiological functions include:

- Transport proteins and their ligands, *e.g.* the sodium-potassium pump, important in the transmission of nerve signals, binds sodium and potassium ions and ATP – see https://pdb101.rcsb.org/motm/118
- DNA and RNA binding proteins and DNA and RNA, *e.g.* the TATA-binding protein, important in transcription, binds the TATA box and other transcription factors– see https://pdb101.rcsb.org/motm/67
- Hormone receptors and their hormones, *e.g.* the estrogen receptor, important in cell growth, binds estrogen and DNA – see https://pdb101.rcsb.org/motm/45
- Riboswitches, important in the regulation of transcription or protein synthesis, and small ligands like ions, amino acids or purine bases – see https://pdb101.rcsb.org/motm/130
- Antibodies, important for the working of our immune system, and antigens – see https://pdb101.rcsb.org/motm/21

- Enzymes and their substrates
- Drugs, *e.g.* HIV reverse transcriptase, important in viral reproduction, binds nucleic acids and the drug AZT – see https://pdb101.rcsb.org/motm/33

Example: TATA Binding Protein

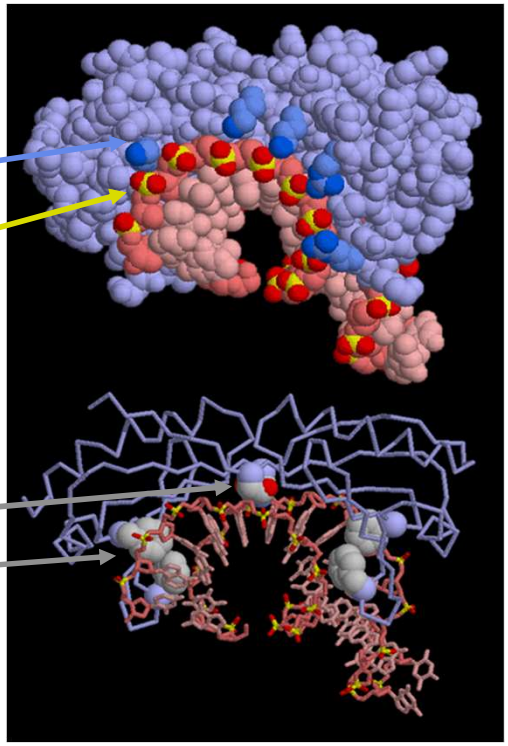


Figure: PDB-101: Molecule of the Month: TATA-Binding Protein

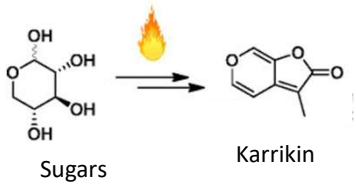
The **TATA Binding Protein (TBP)** is a universal **transcription factor**: it's used in all three eukaryotic RNA polymerases. Its ability to recognize and bind the **TATA box**, a conserved DNA sequence found in many promoters, is a critical first step in transcription.

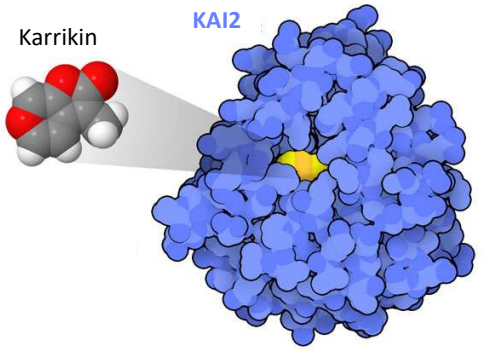
Structurally, TBP has a saddle-like shape: the concave surface binds to the minor groove of DNA. Binding causes a sharp bend (~80°) in the DNA, which helps recruit other transcription factors and RNA polymerase to form the pre-initiation complex.

Basic residues (lysine and arginine) on the binding surface make electrostatic interactions with the negatively charged phosphate backbone. **Phenylalanine** side chains intercalate between base pairs. There are also two symmetrical **asparagine** side acids that form hydrogen bonds at the very centre.

Example: Karrikin









Figures: What are karrikins and how were they 'discovered' by plants? | BMC Biology | Full Text and PDB-101: Molecule of the Month: Phytohormone Receptor DWARF14

Karrikins are fascinating molecules that link environmental events, like wildfires, to biological responses in plants. They are a great example of how chemical signals can trigger developmental processes.

Karrikins are produced during forest fires. When plant material burns, sugars and cellulose undergo chemical transformations, forming karrikins. These molecules are released into the air and settle on the ground in smoke particles. Rain washes karrikins into the soil, where they can enter dormant seeds. Inside the seed, karrikins bind to a receptor protein called **KAI2**. This binding event initiates a signalling cascade that leads to seed germination.

Binding Affinity

Reversible bimolecular reaction: $R + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} RL$

With k_{on} ($M^{-1}\ s^{-1}$) and k_{off} (s^{-1}) the rate constants of the association and dissociation reactions

At equilibrium: effective concentrations of R, L, and RL are constant, but there is flux in each direction: k_{on} [R] [L] = k_{off} [RL]

Equilibrium **dissociation constant** = K_d (M)

$$K_d = \frac{k_{off}}{k_{on}} = \frac{[R][L]}{[RL]}$$

Low K_d → high affinity
High K_d → low affinity

Equilibrium affinity **(association) constant** = K_a (M^{-1})

$$K_a = \frac{k_{on}}{k_{off}} = \frac{[RL]}{[R][L]} = \frac{1}{K_d}$$

The **binding affinity** describes the strength of binding between a receptor and its ligand. Binding affinity is typically measured and reported by the equilibrium **dissociation constant (K_d)**, the reverse of the equilibrium **association constant (K_a)**. K_d can be calculated from k_{on} and k_{off}, the kinetic rate constants that account for the forward binding and reverse unbinding (or dissociation) reaction. Alternatively, K_d can be calculated from the concentrations of receptor, ligand, and receptor-ligand complex at equilibrium.

The smaller the K_d value, the higher the binding affinity, the faster the ligand will bind to the receptor and the longer the receptor-ligand complex will last. The larger the K_d value, the lower the binding affinity, the slower the ligand will bind to the receptor and the quicker the receptor-ligand complex will dissociate.

Knowing the binding affinity is important in research. For example, in **drug design**, the binding affinity is used to identify drug candidates. In general, the higher the affinity of a drug to its target, the better, since lower concentrations of the drug are sufficient to have the desired effect, limiting the risk of adverse side effects.

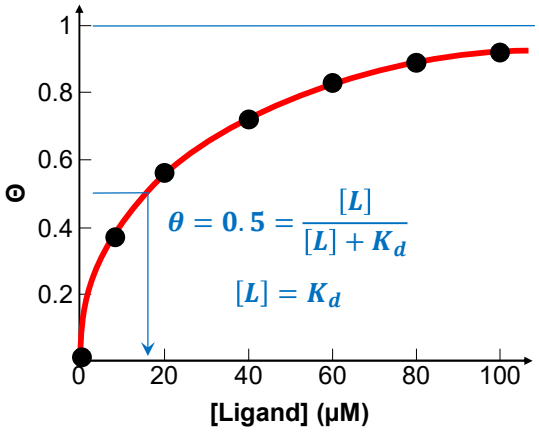
Determining Binding Affinity

Experiment: Vary the concentration of L in a solution containing a fixed, low, concentration of R. At each concentration of ligand, measure **the fraction of bound receptor, θ**.

eq 1 $$\theta = \frac{occupied\ binding\ sites}{total\ binding\ sites} = \frac{[RL]}{[RL] + [R]}$$

To make Θ a function of [L], we use the equilibrium dissociation constant K_d:

eq 2 $$\theta = \frac{\frac{[R][L]}{K_d}}{\frac{[R][L]}{K_d} + [R]} = \frac{\frac{\cancel{[R]}[L]}{\cancel{K_d}}}{\frac{\cancel{[R]}[L]}{\cancel{K_d}} + \frac{\cancel{[R]}K_d}{\cancel{K_d}}} = \frac{[L]}{[L] + K_d}$$



The most common way to determine the K_d value for a given receptor-ligand complex is to vary the concentration of ligand in a solution containing a fixed, low concentration of receptor – constant and much lower than the concentration of ligand used!

At each concentration of ligand ([L]), **the fraction of bound receptor or occupied binding sites (Θ)** – is measured. Θ is a function of [L] and the K_d value. Θ ranges from zero (when [L] = 0) to one (when [L] → ∞). When Θ is ½, [L] is equal to K_d. K_d can therefore be interpreted as the ligand concentration that leads to 50% occupancy of the receptor's binding site. The lower the value of K_d, the less ligand required to achieve 50% occupancy, indicating a higher affinity between receptor and ligand.

Typical Binding Affinity

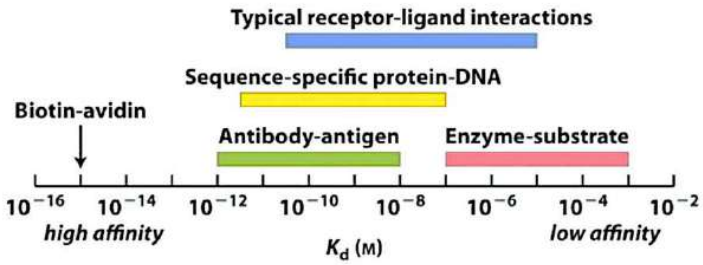
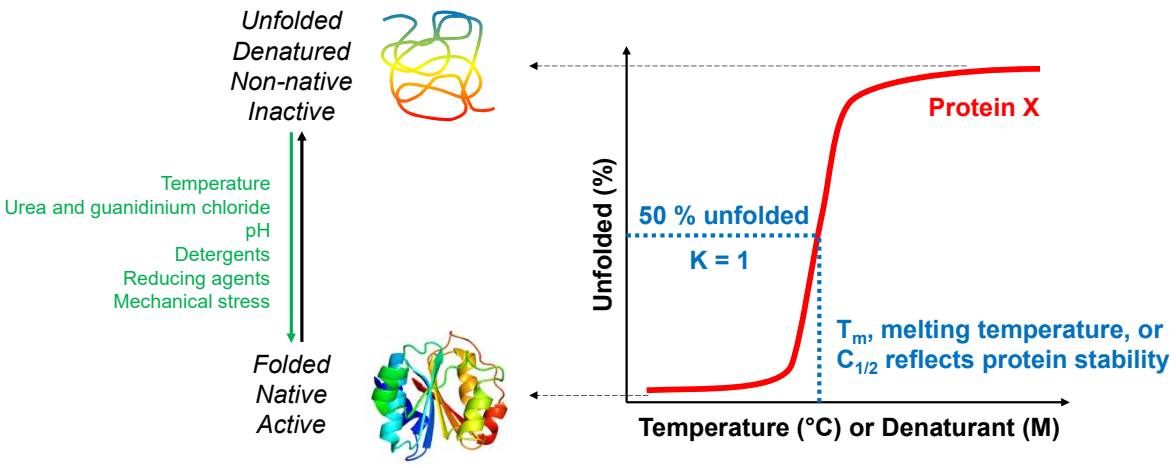


Figure: Lehninger Principles of Biochemistry.

Typical K_d values range from 10^{-15} M to 10^{-3} M.

Protein Unfolding or Denaturation



PDB: 6ADH, horse liver alcohol dehydrogenase

A folded, native, active protein structure is sensitive to environmental factors. A loss of 3D structure sufficient to cause loss of function is called **denaturation or unfolding**. Denaturing conditions include:

- As the **temperature** increases, the rate of molecular vibration increases, and – eventually – weak interactions such as hydrogen bonds are disrupted. As the temperature decreases, ice forms and will break protein structure.
- The **chaotropic agents guanidinium ion and urea**, in concentrations in the range 3 to 10 M, are the most used protein denaturants. They can disrupt hydrophobic interactions.
- **Detergents** contain both hydrophobic and hydrophilic components and can disrupt hydrophobic interactions.
- **Changes in pH** alter the protonation state of certain R groups, which in turn alters hydrogen bonding and salt bridge patterns.
- **Reducing agents** such as β-mercaptoethanol and dithiothreitol convert disulfide bridges to sulfhydryl groups.
- **Mechanical stress** such as stirring and grinding disrupt the delicate balance of forces that maintain protein structure.

Protein stability is a major issue for researchers and industry. For example, antibody instability can lead to reduced production yields and loss of function during storage, a problem for antibody-based drug therapies. There are several ways to measure the stability of proteins, such as **thermal and denaturant resistance**.

Studying the degree of unfolding when the temperature or the concentration of denaturant increases has revealed – for some proteins, not all! – a relatively sharp transition from the folded, or native, form to the unfolded, or denatured, form, depicted in a **denaturation curve**.

Under conditions corresponding to the middle of the transition between the folded and unfolded forms, the protein is "**half folded**". The solution will contain no half-folded molecules but, instead, will be a 50/50 mixture of fully folded and fully unfolded molecules. This means that [Unfolded] = [Folded] and that K = 1.

We determine the temperature at which the protein is 50% unfolded (**T_m or melting temperature**) or the denaturant concentration at which the protein is 50% unfolded (**$C_{1/2}$**).
Most proteins have

- T_m well below 100°C. Among the exceptions are the proteins of thermophilic bacteria, organisms that inhabit hot springs or submarine volcanic vents with temperatures near 100°C.
- $C_{1/2}$ around 5 M urea and 3 M guanidinium chloride.

Anfinsen's Dogma

Proteins can fold spontaneously into their native conformations under physiological conditions.

A protein's primary structure dictates its three-dimensional structure.

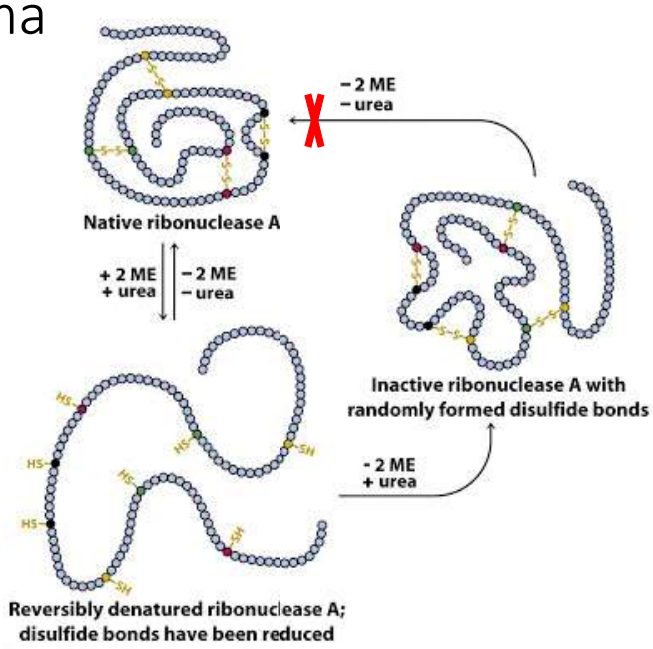


Figure: Horton, Moran, Scrimgeour, Perry, & Rawn (2006), Principles of Biochemistry, Pearson Education

Anfinsen denatured **ribonuclease A (RNase A)**, an enzyme with a single polypeptide chain containing four disulphide bridges, with the **denaturant urea** and **β-mercaptoethanol** (2ME), a **reducing agent** that breaks disulphide bridges.

Careful removal of urea and 2ME with dialysis resulted in a spontaneous and correct refolding of the polypeptide, re-formation of the disulphide bonds, and a full restoration of the enzyme's catalytic activity. The refolding of ribonuclease is so accurate that the four intra-chain disulphide bonds are re-formed in the same positions in the renatured molecule as in the native ribonuclease.

When Anfinsen removed 2ME first and urea second, the protein only regained 1% of its original activity. This result is attributed to the fact that removing 2ME when the enzyme is still unfolded causes cysteine residues to randomly form bonds with each other, indicating that weak bonding interactions are required for correct positioning of disulphide bonds and assumption of the native conformation.

This experiment provided the first evidence that the amino acid sequence of a polypeptide chain contains all the information required to fold the chain into its native, 3D structure. The idea that the amino acid sequence acts as the "blueprint" for protein folding was later referred to as **Anfinsen's dogma**.

You Need to Be Able to...

- ❑ list and recognize some common super-secondary structures
- ❑ define and represent protein tertiary and quaternary structure using topology diagrams
- ❑ look up information in the PDBsum database
- ❑ understand how tertiary structure allows proteins to be flexible and the importance of this flexibility
- ❑ discuss the importance and quantitative aspects of protein-ligand binding
- ❑ determine and evaluate dissociation constants using “fraction of bound receptor” versus “[Ligand]” curves
- ❑ understand and apply the relationship between equilibrium constant and (1) the concentrations of each species in the reaction, (2) forward and reverse rate constants, and (3) the free energy change
- ❑ list conditions leading to protein denaturation
- ❑ use denaturation curves to determine protein stability with T_m or $C_{1/2}$
- ❑ outline the importance of Anfinsen’s work

More Information?

- Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox
  - 4.3 Protein Tertiary and Quaternary Structure
  - 4.4 Protein Denaturation and Folding
  - 5.1 Reversible Binding of a Protein to a Ligand: Oxygen-Binding Proteins
  - 12.4 Receptor Tyrosine Kinases
- Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer
  - 2.5 Tertiary Structure: Proteins Can Fold into Globular or Fibrous Structures
  - 2.6 Quaternary Structure: Polypeptide Chains Can Assemble into Multisubunit Structures
  - 2.7 The Amino Acid Sequence of a Protein Determined Its Three-Dimensional Structure
  - Chapter 3: Binding and Molecular Recognition
  - 14.3 Insulin Signalling: Phosphorylation Cascades Are Central to Many Signal-Transduction Processes